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Inventor(s)	Toba; Shinjiro et al.

Developer container and manufacturing method for developer container

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Abstract

A developer container includes a container member, a connector member attached to the container member, and a discharge path forming member connected to the container member and to discharge a developer from the container member. The connector member includes an inserted portion, and an engaged portion. The discharge path forming member includes an insertion portion inserted into the inserted portion, and an engaging portion provided correspondingly to the engaged portion. The engaged portion and engaging portion come to a state of mutual engagement by a relative rotation between the connector member and the discharge path forming member about a predetermined rotation axis, in a state where the insertion portion is inserted into the inserted portion. In the engaged state, the engaged portion and engaging portion restrict a relative movement of the connector member and the discharge path forming member in the direction of the rotation axis.

Inventors: **Toba; Shinjiro (Kanagawa, JP), Takarada; Hiroshi (Kanagawa, JP)**

Applicant: **CANON KABUSHIKI KAISHA** (Tokyo, JP)

Family ID: **1000008782870**

Assignee: **Canon Kabushiki Kaisha (Tokyo, JP)**

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Primary Examiner: Tran; Hoan H

Attorney, Agent or Firm: Venable LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of International Patent Application No. PCT/JP2022/042455, filed Nov. 15, 2022, which claims the benefit of Japanese Patent Applications No. 2021-198141, filed Dec. 6, 2021, which is hereby incorporated by reference herein in their entirety.

BACKGROUND OF THE INVENTION
Field of the Invention
(1) The present invention relates to a developer container for use in an image forming apparatus that forms images on recording materials.
Background Art
(2) A configuration for suppling developer to a developer container in a main body of an electrophotographic image forming apparatus is known, in which a refill pack as the developer container is attached to the main body of the apparatus, and toner as the developer is supplied from the refill pack to a container portion of the main body of the apparatus (PTL 1). A known refill pack configuration includes a flexible container member (pouch) having an opening portion, to which a discharge path forming member forming a discharge path for discharging the toner is welded, with a shutter that opens and closes a discharge port of the discharge path forming member being assembled thereto.
(3) During the manufacture of the refill pack configuration described above, the pouch could be filled with the toner after welding the discharge path forming member to the opening portion of the pouch through the discharge path formed in the discharge path forming member, for example. However, in some cases, depending on the structure of the discharge port (e.g., opening width), it may not be easy to fill the toner, or it may be difficult to ensure the efficiency of the toner filling operation. Namely, there is some room for improving the efficiency of filling the pouch with toner while maintaining the airtightness of the pouch containing the toner.
(4) An object of the present invention is to provide a technique that can improve the ease of assembly of developer containers while suppressing developer leakage from the container.
CITATION LIST

Patent Literature

(5) PTL 1 Japanese Patent Application Publication No. 2020-154300

SUMMARY OF THE INVENTION

(6) To achieve the above object, a developer container according to the present invention includes: a container member including a container portion containing a developer, and an opening portion that opens the container portion; a connector member attached to the opening portion; and a discharge path forming member connected to the container member via the connector member and forming a discharge path for discharging the developer from the container portion, the connector member including an inserted portion having a through hole extending therethrough, and an engaged portion, the discharge path forming member including an insertion portion inserted into the through hole of the inserted portion, and an engaging portion provided correspondingly to the engaged portion, the engaged portion and the engaging portion being configured to come to a mutually engaged state through a process in which at least one of the connector member and the discharge path forming member is elastically deformed by a relative rotation between the connector member and the discharge path forming member about a predetermined rotation axis as a center axis, in a state where the insertion portion is inserted into the inserted portion, and to restrict, in the engaged state, a relative movement of the connector member and the discharge path forming member in a direction of the rotation axis.

(7) To achieve the above object, a manufacturing method for a developer container of the present invention is a method of manufacturing a developer container that contains a developer to be supplied to a main body of an image forming apparatus, the developer container including: a container member including a container portion containing a developer, and an opening portion that opens the container portion; a connector member attached to the opening portion; and a discharge path forming member connected to the container member via the connector member and forming a discharge path for discharging a developer from the container portion, the connector member including an inserted portion having a through hole extending therethrough, and an engaged portion, the discharge path forming member including an insertion portion inserted into the through hole of the inserted portion, and an engaging portion provided correspondingly to the engaged portion, the engaged portion and the engaging portion being configured to come to a mutually engaged state through a process in which at least one of the connector member and the discharge path forming member is elastically deformed by a relative rotation between the connector member and the discharge path forming member about a predetermined rotation axis as a center axis, in a state where the insertion portion is inserted into the inserted portion, and to restrict, in the engaged state, a relative movement of the connector member and the discharge path forming member in a direction of the rotation axis, the method including: a first assembling step of attaching the connector member to the connector member; a filling step of filling the container portion with the developer; and

a second assembling step of connecting the discharge path forming member to the container member via the connector member.

(8) To achieve the above object, a developer container according to the present invention includes: a container portion having an opening portion at one end thereof and containing developer; a nozzle having an engaging portion, and a discharge port aligned with the container portion in a predetermined direction for discharging the developer contained in the container portion; and a coupling member having an engaged portion that engages with the engaging portion, and attached to the opening portion, and coupling the nozzle and the container portion, the engaged portion including a first restricting portion and a second restricting portion, the engaging portion being configured to move, in a case where the nozzle is moved in an intersecting direction intersecting the predetermined direction relative to the coupling member, from a first position to a second position in the intersecting direction relative to the engaged portion, the first position being a position where the nozzle is allowed to move in the predetermined direction relative to the coupling

member, the second position being a position where the first restricting portion restricts a movement of the nozzle in the predetermined direction relative to the coupling member, and where the second restricting portion restricts a movement of the engaging portion relative to the engaged portion from the second position to the first position in the intersecting direction.
(9) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A is a schematic view illustrating an image forming apparatus according to a first embodiment, and FIG. **1**B is a perspective view illustrating the image forming apparatus.
(2) FIG. **2** is a perspective view illustrating an opening/closing member and a refill port.
(3) FIG. **3**A is an exploded perspective view of a mounting portion, and FIG. **3**B is an exploded perspective view of the mounting portion seen from a different direction than in FIG. **3**A.
(4) FIG. **4**A is a perspective view illustrating an outer view of the mounting portion when the operation lever is in a closed position, and FIG. **4**B is a perspective view illustrating an outer view of the mounting portion when the operation lever is in an open position.
(5) FIG. **5**A is a plan view illustrating an outer view of the mounting portion when the operation lever is in a closed position, and FIG. **5**B is a plan view illustrating an outer view of the mounting portion when the operation lever is in an open position.
(6) FIG. **6**A is a perspective view of an apparatus-side shutter seen from upstream in the mounting direction, and FIG. **6**B is a perspective view of the apparatus-side shutter seen from a different point of view than in FIG. **6**A.
(7) FIG. **7**A is a perspective view of a cover seen from downstream in the mounting direction, and FIG. **7**B is a perspective view of the cover seen from upstream in the mounting direction M.
(8) FIG. **8**A is a cross-sectional view illustrating the mounting portion, and FIG. **8**B is a cross-sectional view illustrating section **8**B-**8**B of FIG. **8**A.
(9) FIG. **9**A is a side view of a toner pack when the pack-side shutter is in a covering position, and FIG. **9**B is a side view of the toner pack when the pack-side shutter is in an open position.
(10) FIG. **10** is an exploded perspective view illustrating the toner pack when the pack-side shutter is in the covering position.
(11) FIG. **11**A is an enlarged perspective view illustrating the vicinity of a nozzle when the pack-side shutter is in the covering position, and FIG. **11**B is a view of the toner pack seen in a removal direction.
(12) FIG. **12**A is an enlarged perspective view illustrating the vicinity of the nozzle when the pack-side shutter is in the open position, and FIG. **12**B is a view of the toner pack seen in the removal direction U.
(13) FIG. **13** is an enlarged perspective view illustrating the vicinity of the nozzle.
(14) FIG. **14** is a side view illustrating the nozzle and the pack-side shutter.
(15) FIG. **15**A is a front view illustrating a pawl, and FIG. **15**B is a cross-sectional view illustrating section **15**B-**15**B of FIG. **15**A.
(16) FIG. **16**A is a front view illustrating the pawl, and FIG. **16**B is a cross-sectional view illustrating section **16**B-**16**B of FIG. **16**A.
(17) FIG. **17**A is a perspective view illustrating the toner pack at one point during the process of being attached to the mounting portion, and FIG. **17**B is a perspective view from a different angle of the toner pack at one point during the process of being attached to the mounting portion.
(18) FIG. **18**A is a cross-sectional view illustrating the toner pack in the process of being attached to the mounting portion, and FIG. **18**B is a cross-sectional view illustrating the toner pack when the

attachment to the mounting portion is completed.
(19) FIG. **19**A is a cross-sectional view illustrating section **19**A-**19**A of FIG. **18**A, and FIG. **19**B is a cross-sectional view illustrating section **19**B-**19**B of FIG. **18**A.
(20) FIG. **20**A is a cross-sectional view illustrating section **20**A-**20**A of FIG. **18**B, and FIG. **20**B is a cross-sectional view illustrating section **20**B-**20**B of FIG. **20**A.
(21) FIG. **21**A is a perspective view illustrating the toner pack in the process of being attached to the apparatus-side shutter, and FIG. **21**B is a cross-sectional view of section **16**B-**16**B of FIG. **16**A illustrating the toner pack when the attachment to the mounting portion is completed.
(22) FIG. **22**A is a perspective view illustrating the operation lever in the closed position and the toner pack, and FIG. **22**B is a perspective view illustrating the operation lever in the open position and the toner pack.
(23) FIG. **23**A is a cross-sectional view illustrating the toner pack and mounting portion when the apparatus-side shutter and pack-side shutter are both in a covering position, and FIG. **23**B is a cross-sectional view illustrating the toner pack and mounting portion when the apparatus-side shutter and pack-side shutter are both in an open position.
(24) FIG. **24**A is a perspective view of a nozzle body of the toner pack according to the embodiment, and FIG. **24**B is a perspective view of the nozzle body of the toner pack according to the embodiment seen from a different point of view than in FIG. **24**A.
(25) FIG. **25**A is a view (plan view) of the nozzle body of the toner pack according to the embodiment seen from the opposite side from the side on which the nozzle body is inserted to and removed from a connector member, and FIG. **25**B is a view (bottom view) of the nozzle body of the toner pack according to the embodiment seen from the side on which the nozzle body is inserted to and removed from the connector member.
(26) FIG. **26**A is a perspective view of the connector member of the toner pack according to the embodiment, and FIG. **26**B is a perspective view of the connector member of the toner pack according to the embodiment seen from a different point of view than in FIG. **26**A.
(27) FIG. **27**A is a perspective view of a pouch and the connector member separate from each other before the connector member is joined to the pouch, FIG. **27**B is a perspective view after the connector member has been joined to the pouch, and FIG. **27**C is a perspective view illustrating a state of the pouch ready to be filled with the toner, with the connector member joined thereto.
(28) FIG. **28**A is an enlarged perspective view of the pouch and nozzle in a state in which the nozzle body is in a fully inserted position relative to the connector member, and FIG. **28**B is an enlarged perspective view of the pouch and nozzle assembly in a state in which the nozzle body is rotated from the fully inserted position shown in FIG. **28**A to a fully engaged position.
(29) FIG. **29**A is a side view illustrating one side of the nozzle body where a discharge port is provided, FIG. **29**B is a cross-sectional view of the connector member including the rotation axis of relative rotation between the nozzle body and connector member, FIG. **29**C is a side view illustrating the opposite side from the side shown in FIG. **29**A of the nozzle body, and FIG. **29**D is a cross-sectional view of the connector member including the rotation axis of relative rotation between the nozzle body and connector member,
(30) FIG. **30**A is an enlarged perspective view illustrating a configuration of an inner engagement protrusion, and FIG. **30**B is an enlarged perspective view illustrating a configuration of an engaged groove.
(31) FIG. **31**A is a schematic enlarged perspective view illustrating the inside of an opening in the connector member that is attached to the pouch, with only the inner engaging protrusion of the nozzle body being shown, when the nozzle body is in the fully inserted position relative to the connector member, FIG. **31**B is a cross-sectional view of section **31**B-**31**B of FIG. **31**A, FIG. **31**C is a schematic enlarged perspective view illustrating the inside of the opening in the connector member that is attached to the pouch, with only the inner engaging protrusion of the nozzle body being shown, when the nozzle body is in the fully engaged position relative to the connector

member, and FIG. **31**D is a cross-sectional view of section **31**D-**31**D of FIG. **31**C.
(32) FIG. **32**A is an enlarged side view of the pouch and nozzle illustrating a state of a second engagement structure (an outer engaging protrusion of the nozzle body and an engaged protrusion of the connector member) when the nozzle body is in the fully inserted position, and FIG. **32**B is a view (bottom view) of the pouch and nozzle shown in FIG. **32**A seen in the insertion direction, FIG. **32**C is an enlarged side view of the pouch and nozzle assembly illustrating a state of the second engagement structure (outer engaging protrusion of the nozzle body and engaged protrusion of the connector member) when the nozzle body is in the fully engaged position, and FIG. **32**D is a view (bottom view) of the pouch and nozzle assembly shown in FIG. **32**C seen in the insertion direction.
(33) FIG. **33**A is an enlarged perspective view illustrating a state of the second engagement structure (outer engaging protrusion of the nozzle body and engaged protrusion of the connector member) when the nozzle body is in the fully engaged position, and FIG. **33**B is an enlarged side view illustrating a state of the second engagement structure (outer engaging protrusion of the nozzle body and engaged protrusion of the connector member) when the nozzle body is in the fully engaged position.
(34) FIG. **34**A is a cross-sectional perspective view of the pouch and nozzle assembly in the vicinity of the nozzle, showing section **34**A-**34**A of FIG. **34**B, and FIG. **34**B is a cross-sectional view of the configuration of a seal member, illustrating section **34**B-**34**B of FIG. **34**A.
(35) FIG. **35**A is a side view illustrating one side of the nozzle body where the discharge port is provided in a variation example, and FIG. **35**B is a cross-sectional view of the connector member including the rotation axis of relative rotation between the nozzle body and connector member, FIG. **35**C is a side view illustrating the opposite side from the side shown in FIG. **35**A of the nozzle body in the variation example, and FIG. **35**D is a cross-sectional view of the connector member including the rotation axis of relative rotation between the nozzle body and connector member.
(36) FIG. **36**A is a schematic enlarged perspective view illustrating the inside of the opening in the connector member that is attached to the pouch, with only the inner engaging protrusion of the nozzle body being shown, when the nozzle body is in the fully inserted position relative to the connector member in the variation example, FIG. **36**B is a cross-sectional view of section **36**B-**36**B of FIG. **36**A, FIG. **36**C is a schematic enlarged perspective view illustrating the inside of the opening in the connector member that is attached to the pouch, with only the inner engaging protrusion of the nozzle body being shown, when the nozzle body is in the fully engaged position relative to the connector member in the variation example, and FIG. **36**D is a cross-sectional view of section **36**D-**36**D of FIG. **36**C.
DESCRIPTION OF THE EMBODIMENTS
(37) The embodiment of the present disclosure will be illustratively described with respect to the following examples of embodiment. It should be noted that the configurations disclosed in the following examples, e.g., functions, materials, shapes of components, and their relative arrangements, are examples in the forms associated with the claims, and are not intended to limit the scope of the claims to the configurations disclosed in these examples. The issues resolved by the configurations disclosed in the following examples, or effects or advantages achieved by the disclosed configurations are not intended to limit the scope of the claims.
Embodiment 1
(38) Hereinafter, an electrophotographic image forming apparatus according to Embodiment 1 of the present disclosure is described with reference to the drawings. An electrophotographic image forming apparatus (hereinafter, image forming apparatus) is an apparatus that uses electrophotographic image forming techniques to form images on recording materials. Examples of image forming apparatus include a copier, facsimile, printer (e.g., laser beam printer, LED printer), and multifunction machine consolidating all these functionalities (multifunction printer).
(39) FIG. **1**A is a schematic view illustrating a configuration of the image forming apparatus **1**

according to this embodiment. FIG. **1**B is a perspective view illustrating the configuration of the image forming apparatus **1**. FIG. **2** is a perspective view illustrating an opening/closing member **83** and a refill port **32***a*.
(40) The image forming apparatus **1** is a monochrome printer that forms images on recording materials P based on image information input from an external device. Recording materials P include various different sheet materials, e.g., paper such as ordinary paper and heavy paper, plastic films such as overhead projector sheet, and irregular-shape sheets such as envelops and index paper, and fabrics.
(41) [Overall Configuration]
(42) The image forming apparatus **1** includes, as shown in FIG. **1**A and FIG. **1**B, an apparatus body **400**, a reading apparatus **200** supported on the apparatus body **400** so as to be openable and closable, and an operation portion **300** attached to an external surface of the apparatus body **400**. The apparatus body **400** includes an image forming portion **10** that forms a toner image on a recording material, a paper feed portion **60** that feeds the recording material to the image forming portion **10**, a fixing portion **70** that fixes the toner image formed in the image forming portion **10** on the recording material, and a pair of discharge rollers **80**.
(43) The image forming portion **10** includes a scanner unit **11**, an electrophotographic processing unit **20**, and a transfer roller **12** that transfers the toner image formed on a photosensitive drum **21** of the processing unit **20** onto the recording material. The processing unit **20** includes the photosensitive drum **21**, a charging roller **22** disposed near the photosensitive drum **21**, a pre-exposure apparatus **23**, and a developing apparatus **30** that includes a developing roller **31**.
(44) The photosensitive drum **21** is a photosensitive member in a cylindrical form. The photosensitive drum **21** in this embodiment has a photosensitive layer made of an organic photosensitive materiel having negative chargeability on a drum-like substrate made of aluminum. The photosensitive drum **21** is rotated at a predetermined processing speed by a motor in a predetermined direction (clockwise in the drawing).
(45) The charging roller **22** contacts the photosensitive drum **21** with a predetermined pressure and forms a charging portion. A charging voltage of a desired level is applied to the charging roller **22** by a high voltage charging power supply, to charge the surface of the photosensitive drum **21** uniformly to a predetermined potential. In this embodiment, the photosensitive drum **21** is charged to a negative polarity by the charging roller **22**. The pre-exposure apparatus **23** reduces the surface potential of the photosensitive drum **21** before the charging portion to ensure a stable discharge in the charging portion.
(46) The scanner unit **11** projects a laser beam corresponding to image information input from an external device or reading apparatus **200** to the photosensitive drum **21** using a polygon mirror to scan and expose the surface of the photosensitive drum **21**. An electrostatic latent image corresponding to the image information is formed on the surface of the photosensitive drum **21** by this exposure. The scanner unit **11** is not limited to a laser scanner device. For example, an LED exposure apparatus having LED array of multiple LEDs aligned along a longitudinal direction of the photosensitive drum **21**, for example, may be adopted.
(47) The developing apparatus **30** includes the developing roller **31** that carries the developer, a developer container **32** that is the casing of the developing apparatus **30**, and a supply roller **33** adapted to supply the developer to the developing roller **31**. The developing roller **31** and supply roller **33** are rotatably supported by the developer container **32**. The developing roller **31** is disposed in an opening portion of the developer container **32** such as to face the photosensitive drum **21**. The supply roller **33** is in contact with and rotatable relative to the developing roller **31**, so that the toner, which is the content stored in the developer container **32**, is applied on the surface of the developing roller **31** by the supply roller **33**. In a configuration in which the toner can be sufficiently supplied to the developing roller **31**, the supply roller **33** need not necessarily be provided.

(48) The developing apparatus **30** of this embodiment uses a contact development system as the developing method. Namely, a toner layer carried on the developing roller **31** comes into contact with the photosensitive drum **21** in a developing portion (developing region) where the photosensitive drum **21** faces the developing roller **31**. A development voltage is applied to the developing roller **31** by a high voltage development power supply. With the development voltage applied, the toner carried on the developing roller **31** is transferred from the developing roller **31** to the drum surface in accordance with the surface potential distribution of the photosensitive drum **21**. The electrostatic latent image is thus developed into a toner image. This embodiment adopts a reversal development method. Namely, the toner image is formed by the toner adhering to a surface region of the photosensitive drum **21** where the charge has decayed by the exposure in the exposure process, after being charged in the charging process.
(49) This embodiment uses a toner having a particle size of 6 [μm], its standard polarity when charged being negative. This embodiment adopts a polymerized toner produced by a polymerization method as one example. The toner in this embodiment is a type known as a nonmagnetic mono-component developer, i.e., the toner does not contain magnetic components, and is carried on the developing roller **31** mainly by an intermolecular force or electrostatic force (image force). Mono-component developer that contains a magnetic component may also be used. Some mono-component developers contain, other than the toner particles, additives (e.g., wax or fine silica particle) for adjusting the fluidity or chargeability of the toner. A dual-component developer composed of a nonmagnetic toner and a magnetic carrier may also be used as the developer. In the case where a magnetic developer is used, a cylindrical developing sleeve with a magnet disposed inside, for example, is used as a developer carrying member.
(50) The developer container **32** is provided with a container portion **36** that contains the toner, and an agitation member **34** disposed inside the container portion **36**. The agitation member **34** is rotated, driven by a motor (not shown), to agitate the toner inside the developer container **32**, as well as to feed the toner to the developing roller **31** and supply roller **33**. The agitation member **34** also serves the function of keeping the toner inside the developer container uniform, by circulating the toner that was not used in the development process and scraped off from the developing roller **31**. The agitation member **34** is not limited to the rotary type. For example, a rocking type agitation member may be adopted.
(51) A developing blade **35**, which restricts the amount of toner carried on the developing roller **31**, is provided in the opening portion of the developer container **32** where the developing roller **31** is disposed. As the developing roller **31** rotates, the toner supplied to the surface of the developing roller **31** is uniformly distributed in a thin layer and negatively charged by triboelectric charging as the toner passes through the area opposite the developing blade **35**.
(52) The paper feed portion **60** includes, as shown in FIG. **1**A and FIG. **1**B, a front door **61** supported on the apparatus body **400** so as to be openable and closable, a tray portion **62**, a middle plate **63**, a tray spring **64**, and a pick-up roller **65**. The tray portion **62** forms the bottom surface of a recording material storage space that is revealed when the front door **61** is opened. The middle plate **63** is supported on the tray portion **62** such as to be lifted up and down. The tray spring **64** urges the middle plate **63** upward to press the recording material P stacked on the middle plate **63** against the pick-up roller **65**. When closed on the apparatus body **400**, the front door **61** closes the recording material storage space. When open relative to the apparatus body **400**, the front door supports the recording material P together with the tray portion **62** and middle plate **63**.
(53) The fixing portion **70** is a heat fixing type that fixes images by heating and melting the toner on the recording material. The fixing portion **70** includes a fixing film **71**, a fixing heater such as a ceramic heater for heating the fixing film **71**, a thermistor for measuring the temperature of the fixing heater, and a pressure roller **72** for making pressure contact with the fixing film **71**.
(54) Next, an image forming operation of the image forming apparatus **1** is described. An input of an image formation command to the image forming apparatus **1** initiates the image forming process

by the image forming portion **10** based on the image information input from an external computer connected to the image forming apparatus **1**, or from the reading apparatus **200**. The scanner unit **11** projects a laser beam to the photosensitive drum **21** in accordance with the input image information. The photosensitive drum **21** at this time is already charged by the charging roller **22**, so that an electrostatic latent image is formed on the photosensitive drum **21** by the laser beam irradiation. The developing roller **31** then develops this electrostatic latent image and thus a toner image is formed on the photosensitive drum **21**.
(55) In parallel with the above image forming process, the pick-up roller **65** in the paper feed portion **60** rolls out the recording material P supported on the front door **61**, tray portion **62**, and middle plate **63**. The recording material P, fed to a pair of registration rollers **15** by the pick-up roller **65**, abuts on the nip of the registration roller pair **15**, whereby any skew is corrected. The pair of registration rollers **15** is driven in sync with the transfer timing of the toner image, to convey the recording material P toward a transfer nip formed by the transfer roller **12** and the photosensitive drum **21**.
(56) A transfer voltage is applied to the transfer roller **12** by a high voltage transfer power supply, so that the toner image carried on the photosensitive drum **21** is transferred onto the recording material P as it is conveyed by the pair of registration rollers **15**. The recording material P onto which the toner image has been transferred is conveyed to the fixing portion **70**, where the toner image is heated and pressed as the recording material passes through a nip portion between the fixing film **71** and the pressure roller **72** of the fixing portion **70**. The heat and pressure melt the toner particles, which then adhere, and the toner image is thus fixed to the recording material P. The recording material P, after passing through the fixing portion **70**, is expelled from the image forming apparatus **1** (to the outside) by a pair of discharge rollers **80**, and stacked on a discharge tray **81** formed in a top part of the apparatus body **400**.
(57) The discharge tray **81** is inclined upward toward downstream in the discharge direction of the recording material. The recording material expelled onto the discharge tray **81** slides down on the discharge tray **81**, whereby the rear edges are aligned by a restricting surface **84**.
(58) The reading apparatus **200** includes a reading unit **201** containing a reading portion (not shown) inside, and a pressing plate **202** supported on the reading unit **201** so as to be openable and closable. A document glass **203**, which transmits the light emitted from the reading portion, and on which a document is placed, is provided on the upper surface of the reading unit **201**.
(59) For an image of a document to be read by the reading apparatus **200**, the user places the document on the document glass **203**, with the pressing plate **202** open. The pressing plate **202**, when shut, prevents misalignment of the document on the document glass **203**. A read command is output to the image forming apparatus **1** by an operation performed to the operation portion **300**, for example. The reading operation is started with the reading portion inside the reading unit **201** moving back and forth in a sub-scanning direction, i.e., left and right direction, with the operation portion **300** of the image forming apparatus **1** positioned in the front. The reading portion reads the image of the document by photoelectric conversion, i.e., by emitting light from a light-emitting portion to the document and receiving the light reflected from the document by a light-receiving portion. Hereinafter, the front and back direction, left and right direction, and up and down direction are defined relative to the operation portion **300** facing the front.
(60) A top cover **82** is provided in an upper part of the apparatus body **400**. The discharge tray **81** is formed on the upper surface of the top cover **82**. An opening/closing member **83** is supported on the top cover **82** rotatably about a rotation shaft **83***a* extending in the front and back direction to open and close, as shown in FIG. **1**B and FIG. **2**. An opening portion **82***a* that opens upward is formed in the discharge tray **81** of the top cover **82**.
(61) The opening/closing member **83** is configured to be movable between a closed position where it covers the refill port **32***a* so as not to allow attachment of a toner pack **100** to the developer container **32**, and an open position where it exposes the refill port **32***a* to allow the toner pack **100**

to be attached to the developer container **32**. With the opening/closing member **83** being in the open position, the toner pack **100** is moved in the mounting direction M toward the refill port **32***a*, and attached to the refill port **32***a*.
(62) In the closed position, the opening/closing member **83** serves as a part of the discharge tray **81**. The opening/closing member **83** and opening portion **82***a* are formed on the left side of the discharge tray **81**. The opening/closing member **83** is opened to the left by hooking a finger in a groove **82***b* provided in the top cover **82**. The opening/closing member **83** is substantially in the form of letter L conforming to the shape of the top cover **82**.
(63) The opening portion **82***a* of the discharge tray **81** is opened to expose the refill port **32***a* formed at the top of the developer container **32** for suppling the toner. The user can access the refill port **32***a* by opening the opening/closing member **83**. This embodiment adopts a direct refill system that allows the user to supply the toner from the toner pack **100** filled with refill toner (see FIG. **1**A and FIG. **1**B) to the developing apparatus **30**, with the developing apparatus **30** being mounted in the image forming apparatus **1**. The toner pack **100** is at least partly exposed to the outside when attached to a mounting portion **106** of the image forming apparatus **1** (see FIG. **17**A and FIG. **17**B).
(64) This system offers better usability because it obviates the need to take the processing unit **20** out of the apparatus body **400** and to replace it with a new one, when the processing unit **20** runs low on the toner. This system also allows the developer container **32** to be refilled with toner more inexpensively than replacing the entire processing unit **20**. The direct refill system also enables cost reduction as compared to replacing only the developing apparatus **30** of the processing unit **20**, because there is no need to replace various rollers and gears. The image forming apparatus **1** and the toner pack **100** configure an image forming system **1000**.
(65) [Mounting Portion]
(66) Next, the configuration of the mounting portion **106** where the toner pack **100** is attached is described with reference to FIG. **3**A to FIG. **8**B. In this embodiment, the mounting portion **106** is a unit including the refill port **32***a*, provided to the image forming apparatus **1** (see FIG. **2**) for allowing attachment of the toner pack **100**. FIG. **3**A is an exploded perspective view of the mounting portion **106**. FIG. **3**B is an exploded perspective view of the mounting portion **106** seen from a different direction than in FIG. **3**A. FIG. **4**A is a perspective view illustrating an outer view of the mounting portion **106** when the operation lever **108** is in a closed position, and FIG. **5**A is a view of the mounting portion **106** seen from the mounting direction M when the operation lever **108** is in the closed position. FIG. **4**B is a perspective view illustrating an outer view of the mounting portion **106** when the operation lever **108** is in an open position, and FIG. **5**B is a view of the mounting portion **106** seen from the mounting direction M when the operation lever **108** is in the open position.
(67) FIG. **6**A is a perspective view of an apparatus-side shutter **109** seen from upstream in the mounting direction M. FIG. **6**B is a perspective view of the apparatus-side shutter **109** seen from a different point of view than in FIG. **6**A. FIG. **7**A is a perspective view of a cover **110** seen from downstream in the mounting direction M. FIG. **7**B is a perspective view of the cover **110** seen from upstream in the mounting direction M. FIG. **8**A is a cross-sectional view illustrating the mounting portion **106**. FIG. **8**B is a cross-sectional view illustrating section **8**B-**8**B of FIG. **8**A.
(68) As shown in FIG. **3**A to FIG. **4**B, the mounting portion **106** includes a body base portion **2**. The body base portion **2** includes a first frame **107**, a second frame **117**, and a cover **110**. The cover **110** and second frame **117** are fixed to the first frame **107**. As shown in FIG. **7**A and FIG. **7**B, the cover **110** has an engaged portion **110***h* that engages with an engaging portion **107***b* of a positioning portion **107***a* of the first frame **107** so as not to rotate about a rotation axis B relative to the first frame **107**. A notch **110***k* is formed downstream of the cover **110** in the mounting direction M, i.e., on the bottom side. The notch **110***k* has a first restricting surface **110***c* and a second restricting surface **110***d*. The first restricting surface **110***c* and second restricting surface **110***d* are formed opposite each other in the circumferential direction around the rotation axis B.

(69) The first frame **107**, cover **110**, and second frame **117** may be in one piece instead of separate parts. As shown in FIG. **3**A and FIG. **3**B, the second frame **117** is formed with an apparatus-side opening **117***a*. The apparatus-side opening **117***a* communicates with the container portion **36** of the developer container **32** (see FIG. **1**A).
(70) The operation lever **108** and apparatus-side shutter **109** are each mounted to the body base portion **2** such as to be rotatable about the rotation axis B. The first frame **107** is provided with the positioning portion **107***a*. The positioning portion **107***a* protrudes inwards more than the inner circumferential surface **107***c* around the rotation axis B of the first frame **107** in the radial direction r of a virtual circle VC around the rotation axis B.
(71) The operation lever **108** as a part to be operated is provided with a drive-transmitting portion **108***a* and an operation portion **108***b*. The user can turn the operation lever **108** about the rotation axis B relative to the body base portion **2** by operating the operation portion **108***b*. As shown in FIG. **3**A, the drive-transmitting portion **108***a* is a protrusion inwardly protruding more than the inner circumferential surface around the rotation axis B of the operation lever **108** in the radial direction r of the virtual circle VC around the rotation axis B.
(72) The apparatus-side shutter **109** that is the shutter in the main body includes an inner circumferential surface **109***h*, an inlet port **109***a* formed in the inner circumferential surface **109***h* for receiving the toner from the toner pack **100**, and a bottom surface **109***b*, as shown in FIG. **6**A and FIG. **6**B. The apparatus-side shutter **109** further includes a center boss **109***d* formed on the bottom surface **109***b*, a pack contact surface **109***g*, a restriction rib **109***c*, and a drive transmitted portion **109***e* provided on the inner circumferential surface **109***h*. The drive transmitted portion **109***e* is a protrusion inwardly protruding in the radial direction r of the virtual circle VC around the rotation axis B, as shown in FIG. **6**A. An apparatus-side seal **111** is bonded around the inlet port **109***a* on the inner circumferential surface **109***h* (see FIG. **4**B).
(73) The apparatus-side shutter **109** is configured to assume a covering position as a second covering position and an open position as a second open position relative to the body base portion **2**. More specifically, the apparatus-side shutter **109** rotates in the direction of arrow K from the covering position to the open position, and rotates in the direction of arrow L from the open position to the covering position, as shown in FIG. **6**A and FIG. **6**B. These directions of arrow K and arrow L are the same as the directions of arrow K and arrow L of a pack-side shutter **103** shown in FIG. **11**A. The inlet port **109***a* is covered by the apparatus-side seal **111** and cover **110** when the apparatus-side shutter **109** is in the covering position, and the inlet port **109***a* is opened, or not covered by the cover **110**, in the open position. Namely, the inlet port **109***a* does not communicate with the apparatus-side opening **117***a* of the second frame **117** when the apparatus-side shutter **109** is in the covering position, and communicates with the apparatus-side opening **117***a* of the second frame **117** when the apparatus-side shutter **109** is in the covering position.
(74) The apparatus-side shutter **109** is in the covering position in FIG. **4**A and FIG. **5**A. At this time, the inlet port **109***a* of the apparatus-side shutter **109** does not communicate with the apparatus-side opening **117***a* of the second frame **117**. The apparatus-side shutter **109** is in the open position in FIG. **4**B and FIG. **5**B. At this time, the inlet port **109***a* of the apparatus-side shutter **109** communicates with the apparatus-side opening **117***a* of the second frame **117**. The apparatus-side shutter **109** moving to the open position allows the toner to be replenished (supplied) from the toner pack **100** to the container portion **36** of the developer container **32** through the inlet port **109***a*.
(75) The operation lever **108** is not linked to the apparatus-side shutter **109** such as to transmit a drive force. Therefore, if manipulated without the toner pack **100** being attached, the operation lever **108** does not cause the apparatus-side shutter **109** to rotate.
(76) As shown in FIG. **8**A and FIG. **8**B, the apparatus-side shutter **109** is configured to be rotatable about the center boss **109***d*, with a large-diameter portion **109***d***1** of the center boss **109***d* fitting into a cylindrical portion **110***j* of the cover **110**. The restriction rib **109***c* provided on the bottom surface

109*b* of the apparatus-side shutter **109** is located between the first restricting surface **110***c* and the second restricting surface **110***d* of the cover **110**. Therefore, the apparatus-side shutter **109** is rotatable only in the movable range of the restriction rib **109***c* between the first restricting surface **110***c* and the second restricting surface **110***d*. In other words, the rotation of the apparatus-side shutter **109** is restricted to the range between the covering position and open position by the first restricting surface **110***c* and the second restricting surface **110***d* of the cover **110**. For example, when the restriction rib **109***c* is in contact with the first restricting surface **110***c* as shown in FIG. **8**B, the apparatus-side shutter **109** located in the covering position cannot rotate in the direction of arrow L, i.e., in the opposite direction from the direction toward the open position.

(77) [Toner Pack Configuration]

(78) Next, a basic configuration of the toner pack **100** is described with reference to FIG. **9**A to FIG. **10**. The toner pack **100** is attached to the mounting portion **106** described above. FIG. **9**A is a side view of the toner pack **100** when the pack-side shutter **103** is in a covering position. FIG. **9**B is a side view of the toner pack **100** when the pack-side shutter **103** is in an open position. FIG. **10** is an exploded perspective view of the toner pack **100** when the pack-side shutter **103** is in the open position.

(79) The toner pack **100** includes a pouch **101** containing toner, a nozzle **102** joined to the pouch **101**, and the pack-side shutter **103**, as shown in FIG. **9**A to FIG. **10**. The nozzle **102** and pack-side shutter **103** are connected to the pouch **101**, and form a mounted portion **700** attached to the mounting portion **106**.

(80) The pouch **101** as a container member has flexibility. The pouch is provided at one end of the toner pack **100** in the axial direction D**1** that is the direction of a rotation axis A of the pack-side shutter **103**. The rotation axis A coincides with the rotation axis B of the apparatus-side shutter **109** when the toner pack **100** is attached to the mounting portion **106**. Hereinafter, the axial direction of the rotation axis A and rotation axis B will both be referred to as axial direction D**1**. The nozzle **102** and pack-side shutter **103** are provided at the other end of the toner pack **100** in the axial direction D**1**. The pouch **101** is formed by a lamination process from a flexible polypropylene sheet into a bag shape with one end open, for example. The pouch **101** may be a resin bottle, or paper or plastic container.

(81) The pouch **101** as a container member includes a container portion **101***a* containing the toner, and an opening portion **101***b* that opens the container portion **101***a*. The nozzle **102** is joined to the opening portion **101***b* of the pouch **101** such that the container portion **101***a* of the pouch **101** communicates with the outside of the toner pack **100** only through a toner discharge path (see FIG. **23**B) formed by the nozzle **102**. The nozzle **102** is formed by a nozzle body **121**, which is a discharge path forming member, forming a discharge path for the toner to be discharged from the container portion **101***a* of the pouch **101**, and a connector member (coupling member) **122** for attaching the nozzle body **121** to the pouch **101**. The connector member **122** is joined to the opening portion **101***b* of the pouch **101**. The joining method is not limited to a particular method. Examples of joining methods include the use of various adhesives such as a hot-melt adhesive, or joining by heat-bonding the pouch **101** to the outer periphery of the connector member **122**. The nozzle **102** is joined to the pouch **101** by the nozzle body **121** engaging with the connector member **122**. The engagement structure between the nozzle body **121** and the connector member **122** will be described in detail later.

(82) The nozzle **102** has an outer surface extending along the rotation axis A, and a side face **102***c* as a first outer surface. In the side face **102***c* are provided a discharge port **102***a* configured to communicate with the inside of the pouch **101**, and a recessed portion **102***e*. The recessed portion **102***e* is provided at a different position than that of the discharge port **102***a* in the rotating direction of the pack-side shutter **103**. The pouch is configured such that the toner contained in the pouch **101** is discharged to the outside of the toner pack **100** through the discharge port **102***a* when the volume of the pouch **101** is reduced by the user squeezing the pouch **101**. Namely, the nozzle **102**

has a passage **102***g* inside (see FIG. **23**B) as the discharge path for the toner (content) to pass through toward the discharge port **102***a*.

(83) The pack-side shutter **103** as a shutter is disposed on the outer side of the side face **102***c* of the nozzle **102**. The pack-side shutter **103** is rotatable about the rotation axis A extending along the axial direction D**1**, and has an opening **103***a*. Specifically, an inner circumferential surface **103***m* of the pack-side shutter **103** is slidably supported on an annular rib **102***m* of the nozzle **102**. The pack-side shutter **103** is provided on the outer side of the side face **102***c* in the radial direction r of the virtual circle VC around the rotation axis A. The side face **102***c* has an arcuate surface that is a curved surface outwardly convex in the radial direction r. The pack-side shutter **103** has an inner surface, i.e., opposite the side face **102***c*, which is a curved surface conforming to the side face **102***c* of the nozzle **102**, where a substantially rectangular pack-side seal **105** is attached.

(84) The pack-side shutter **103** is configured to be rotatable about the rotation axis A between a covering position where the pack-side seal **105** covers the discharge port **102***a* of the nozzle **102** (position shown in FIG. **9**A) and an open position where the seal opens the discharge port **102***a* (position shown in FIG. **9**B). When the pack-side shutter **103** is in the open position, the opening **103***a* in the pack-side shutter **103** exposes the discharge port **102***a* of the nozzle **102**.

(85) When the pack-side shutter **103** located in the covering position as a first covering position shown in FIG. **9**A is rotated about the rotation axis A in the direction of arrow K, the pack-side shutter **103** reaches the open position as a first open position shown in FIG. **9**B. Conversely, when the pack-side shutter **103** located in the open position is rotated in the direction of arrow L, the pack-side shutter **103** reaches the covering position. Namely, the direction of arrow K as a first rotating direction is the direction from the covering position to the open position around the rotation axis A. The direction of arrow L as a second rotating direction is the direction from the open position to the covering position around the rotation axis A. The pack-side shutter **103** makes frictional contact with the side face **102***c* of the nozzle **102** via the pack-side seal **105** during the rotating motion of the pack-side shutter **103**.

(86) Next, the detailed configurations of the nozzle **102** and the pack-side shutter **103** are described with reference to FIG. **11**A to FIG. **14**. FIG. **11**A is an enlarged perspective view illustrating the vicinity of the nozzle **102** when the pack-side shutter **103** is in the covering position. FIG. **11**B is a view when the toner pack **100** is seen in a removal direction U in FIG. **11**A. FIG. **12**A is an enlarged perspective view illustrating the vicinity of the nozzle **102** when the pack-side shutter **103** is in the open position. FIG. **12**B is a view when the toner pack **100** is seen in the removal direction U in FIG. **12**A. FIG. **13** is an enlarged perspective view illustrating the vicinity of the nozzle **102**. FIG. **14** is a side view illustrating the nozzle **102** and the pack-side shutter **103**. The removal direction U is the opposite direction from the mounting direction M; it is the direction in which the toner pack **100** moves when removed from the mounting portion **106**.

(87) As shown in FIG. **11**A and FIG. **11**B, the nozzle **102** has a positioned portion **102***d* with surfaces **102***d***1** and **102***d***2** spaced apart and aligned in the direction of arrow R, and extending in a direction intersecting the direction of arrow R. As shown in FIG. **11**B, the surfaces **102***d***1** and **102***d***2** in this embodiment extend in a direction perpendicular to the direction of arrow R, and are parallel to each other. Namely, the direction of arrow R in this embodiment is the direction of normal line of the surfaces **102***d***1** and **102***d***2**. The positioned portion **102***d* engages with the positioning portion **107***a* of the first frame **107** (FIG. **4**A) when the toner pack **100** is attached to the mounting portion **106**. This determines the position of the nozzle **102** relative to the first frame **107** (body base portion **2**) in the direction of arrow R (in the rotating direction around the rotation axis A). In FIG. **11**B, a line CL**1** passing through the center between the surface **102***d***1** and surface **102***d***2** aligned in the direction of arrow R and extending perpendicularly to the direction of arrow R is shifted by about 90° from a line CL**2** passing through the rotation axis A and the center of the discharge port **102***a*.

(88) As shown in FIG. **11**A and FIG. **14**, along the direction of the rotation axis A, a surface **102***e***1**

and a surface **102***e***2** are provided downstream in the mounting direction M of the surface **102***d***1** and surface **102***d***2**, respectively. The surface **102***e***1** and surface **102***e***2** extend in the radial direction r of the virtual circle VC around the rotation axis A, as shown in FIG. **11**B. Note, the extending direction of the surface **102***e***1** and surface **102***e***2** is not limited to this example, and may be set in any direction as long as they do not interfere with the positioning portion **107***a* of the first frame **107**.
(89) As shown in FIG. **14**, a side face **102***e***3** is provided between the surface **102***d***1** and the surface **102***d***2**, and between the surface **102***e***1** and the surface **102***e***2** in the direction of arrow R. The side face **102***e***3** is set back inward from the side face **102***c* in the radial direction r. The surfaces **102***d***1**, **102***d***2**, **102***e***1**, and **102***e***2**, and side face **102***e***3** together form the recessed portion **102***e*.
(90) The surface **102***d***1** and surface **102***d***2** need not necessarily be parallel as in this embodiment. For example, the surface **102***d***1** and surface **102***d***2** may be surfaces extending in the radial direction r of the virtual circle VC around the rotation axis A. In this case, the direction of arrow R will be the tangential direction of the virtual circle VC, and the line CL**1** perpendicular to the direction of arrow R can be set at any angle relative to the line CL**2**.
(91) As shown in FIG. **11**A and FIG. **11**B, when viewed in a direction perpendicular to the axial direction D**1** of the rotation axis A, the opening **103***a* is provided in the side face **103***d* of the pack-side shutter **103**. As shown in FIG. **11**A, when the pack-side shutter **103** is in the covering position, the recessed portion **102***e* of the nozzle **102** is at least partly exposed from the opening **103***a*. This is to allow the surface **102***d***1** and surface **102***d***2** of the recessed portion **102***e*, i.e., the positioned portion **102***d*, to engage with the positioning portion **107***a*, when the toner pack **100** is attached to the mounting portion **106**, with the pack-side shutter **103** located in the covering position.
(92) As shown in FIG. **11**B, the pack-side shutter **103** is further provided with a drive transmitted portion **103***e* on the opposite side of the rotation axis A from the opening **103***a*. The drive transmitted portion **103***e* is on the opposite side of the rotation axis A from the recessed portion **102***e* of the nozzle **102** when the pack-side shutter **103** is located in the covering position. The drive transmitted portion **103***e* includes surfaces **103***b***1** and **103***b***2** and a side face **103***b***3**, and can engage with the drive-transmitting portion **108***a* of the operation lever **108** to be described later. The surface **103***b***1** and surface **103***b***2** both extend in a direction perpendicular to the direction of arrow R. FIG. **13** is an enlarged perspective view of the vicinity of the pack-side shutter **103** when viewed from the side with the drive transmitted portion **103***e*. Between the surfaces **103***b***1** and **103***b***2** is provided the side face **103***b***3** that is set back inward from the side face **103***d* in the radial direction r.
(93) A protruded portion **102***b* of the nozzle **102** is described with reference to FIG. **11**A to FIG. **14**. The toner pack **100** is oriented such that the second end of the toner pack **100** (nozzle **102** side) is positioned below the first end (pouch **101** side) as shown in FIG. **9**A and FIG. **9**(*b*). Put differently, the toner pack **100** is oriented such that the nozzle **102** is positioned at least partly below the pouch **101**, with the rotation axis A being parallel to the vertical direction. The toner pack **100** is oriented this way when attached to the mounting portion **106** of the image forming apparatus **1**. The mounting direction M at this time is downward, and the removal direction U upward, in FIG. **11**A and FIG. **12**A.
(94) The pack-side shutter **103** has an end face **103***c* as a shutter end face, which is a lower end face in the vertical direction VD, forming the bottom surface of the pack-side shutter **103**. The nozzle **102** has the protruded portion **102***b* as a first protruding portion, downstream of the end face **103***c* of the pack-side shutter **103** in the mounting direction M, i.e., protruding downward. The protruded portion **102***b* is cylindrical (a portion having a cylindrical shape) having the rotation axis A in the center, as shown in FIG. **11**A. The protruded portion **102***b* has a protruded end face **102***b***2**, which is a lower end face. The protruded end face **102***b***2** is formed with a hole having an inner circumferential surface **102***b***1** around the rotation axis A. The protruded portion **102***b* protrudes downward more than a lower end face **102***j* of the nozzle **102** as shown in FIG. **10**. While the end

face **103***c* of the pack-side shutter **103** and the end face **102***j* of the nozzle **102** are perpendicular to the rotation axis A in this embodiment, the direction of these end faces is not limited to this. These surfaces may extend in any direction intersecting the rotation axis A when viewed from a direction perpendicular to the rotation axis A. The protruded portion **102***b* need not necessarily be provided to the nozzle **102**.

(95) Referring now to FIG. **15**A, the nozzle **102** of the toner pack **100** is provided with a pawl **102***f* as a lock mechanism for preventing rotation of the pack-side shutter **103** relative to the nozzle **102** during transportation or when the user handles the toner pack **100** alone. The pack-side shutter **103** is retained in the covering position by the pawl **102***f*, so that leakage of the toner, or the content of the toner pack **100**, can be prevented.

(96) FIG. **15**A is a front view illustrating the pawl **102***f*. FIG. **15**B is a cross-sectional view illustrating section **15**B-**15**B of FIG. **15**A. FIG. **16**A is a front view illustrating the pawl **102***f*. FIG. **16**B is a cross-sectional view illustrating section **16**B-**16**B of FIG. **16**A.

(97) The pawl **102***f* as a second restricting portion includes an arm **102***f***3**, an unlocking inclined surface **102***f***1**, and an abutting portion **102***f***2**, as shown in FIG. **15**A and FIG. **15**B. The pawl **102***f* is movable in the radial direction r of the virtual circle VC around the rotation axis A by elastic deformation of the arm **102***f***3**. Specifically, the pawl **102***f* can move between a restricting position shown in FIG. **15**B, and a non-restricting position shown in FIG. **21**B that is more inside than the restricting position in the radial direction r.

(98) When the pawl **102***f* is located in the restricting position as shown in FIG. **15**B, the pack-side shutter **103** is located in the covering position, and the abutting portion **102***f***2** is positioned opposite the restricting portion **103***h* in the circumferential direction around the rotation axis A. At this time, there is a gap s between the abutting portion **102***f***2** and the restricting portion **103***h*. The rotation in the direction of arrow K of the pack-side shutter **103** is restricted by the restricting portion **103***h* abutting the abutting portion **102***f***2**. The gap s may be set as suited. The pack-side shutter **103** is deemed to be in the covering position when it is in the rotatable range in which it is allowed to rotate by the gap s. Namely, the pawl **102***f* located in the restricting position restricts the rotation of the pack-side shutter **103** from the covering position in the direction of arrow K.

(99) When the pawl **102***f* is located in the non-restricting position, the abutting portion **102***f***2** is located more inside than the restricting portion **103***h* of the pack-side shutter **103** in the radial direction r of the virtual circle VC around the rotation axis A. Therefore, the pack-side shutter **103** can rotate about the rotation axis A without interfering with the abutting portion **102***f***2**.

(100) As shown in FIG. **6**A and FIG. **21**B, the apparatus-side shutter **109** is provided with an unlocking rib **109***j* extending in the axial direction D**1**. The unlocking rib **109***j* can make contact with the unlocking inclined surface **102***f***1** of the pawl **102***f* when the toner pack **100** is attached to the mounting portion **106**. As shown in FIG. **11**B, FIG. **15**B, FIG. **17**B, and FIG. **21**B, the pack-side shutter **103** is provided with an opening **103***j*, which extends from the end face (bottom surface) **103***c* to the side face **103***d* of the pack-side shutter **103**. The unlocking rib **109***j* provided to the apparatus-side shutter **109** extends through the opening **103***j* and can make contact with the unlocking inclined surface **102***f***1** of the pawl **102***f* that is disposed inside the pack-side shutter **103**.

(101) The unlocking inclined surface **102***f***1** is inclined relative to the mounting direction M (axial direction D**1**) such as to extend inward in the radial direction r toward downstream in the mounting direction M. When the toner pack **100** is attached to the mounting portion **106**, the unlocking inclined surface **102***f***1** directs the force the pawl **102***f* receives from the unlocking rib **109***j* inward in the radial direction r. With the unlocking inclined surface **102***f***1** being pressed by the unlocking rib **109***j*, the pawl **102***f* moves inward in the radial direction r from the restricting position toward the non-restricting position. In other words, the pawl **102***f* moves from the restricting position toward the non-restricting position by being pressed by the mounting portion **106** when the toner pack **100** is attached to the mounting portion **106**.

(102) The abutting portion **102***f***2** of the pawl **102***f* described above restricts the rotation of the pack-

side shutter **103** in the direction of arrow K by abutting on the restricting portion **103***h* of the pack-side shutter **103**. Next, the configuration that restricts the rotation of the pack-side shutter **103** in the direction of arrow L, i.e., opposite from the direction of arrow K, is described.

(103) As shown in FIG. **16**B, the pack-side shutter **103** has a rotation restricting rib **103***k*, while the nozzle **102** has a rotation restricting surface **102***k* that faces the rotation restricting rib **103***k* in the circumferential direction around the rotation axis A. When the pack-side shutter **103** is located in the covering position, the rotation restricting rib **103***k* is opposite the rotation restricting surface **102***k* with a slight gap therebetween. An attempt to rotate the pack-side shutter **103** located in the covering position in the direction of arrow L results in the rotation restricting rib **103***k* abutting the rotation restricting surface **102***k*, thereby stopping the rotation of the pack-side shutter **103** in the direction of arrow L.

(104) As shown in FIG. **16**A and FIG. **16**B, the pawl **102***f* is disposed downstream of the rotation restricting surface **102***k* and rotation restricting rib **103***k* in the mounting direction M. This is because of the pawl **102***f* provided at a position where it can readily be pressed by the unlocking rib **109***j* of the apparatus-side shutter **109** when the toner pack **100** is attached to the mounting portion **106**. This allows the opening **103***j* of the pack-side shutter **103** to be reduced in size, which ensures rigidity of the pack-side shutter **103**, as well as deters user's access to the pawl **102***f*. The arrangement of the pawl **102***f*, rotation restricting surface **102***k* and rotation restricting rib **103***k* is not limited to this example and may be changed as suited.

(105) As described above, when the toner pack **100** is not attached to the mounting portion **106**, the pack-side shutter **103** is restricted from rotating in the direction of arrow K and in the direction of arrow L, and readily retained in the covering position. After the toner pack **100** is attached to the mounting portion **106**, with the pawl **102***f* in the non-restricting position, when the pack-side shutter **103** is rotated in the direction of arrow K, the pack-side shutter **103** exposes the discharge port **102***a* of the nozzle **102** as shown in FIG. **12**A.

(106) As shown in FIG. **11**A and FIG. **13**, the pack-side shutter **103** is provided with three radial positioning portions **103***f*. These radial positioning portions **103***f* protrude outward in the radial direction r more than the side face **103***d*. The radial positioning portions **103***f* are each disposed on the upstream side of the pack-side shutter **103** in the mounting direction M.

(107) [Attachment of Toner Pack to Mounting Portion]

(108) Next, the conditions when the toner pack **100** is attached to the mounting portion **106** are described with reference to FIG. **17**A to FIG. **21**B. FIG. **17**A and FIG. **17**B are perspective views from different angles illustrating a condition at one point during the process of attaching the toner pack **100** to the mounting portion **106**. FIG. **18**A is a cross-sectional view illustrating a condition in the process of attaching the toner pack **100** to the mounting portion **106**. FIG. **18**B is a cross-sectional view illustrating a condition when the attachment of the toner pack **100** to the mounting portion **106** is completed.

(109) FIG. **19**A is a cross-sectional view illustrating section **19**A-**19**A of FIG. **18**A. FIG. **19**B is a cross-sectional view illustrating section **19**B-**19**B of FIG. **18**A. FIG. **20**A is a cross-sectional view illustrating section **20**A-**20**A of FIG. **18**B. FIG. **20**B is a cross-sectional view illustrating section **20**B-**20**B of FIG. **20**A. FIG. **21**A is a perspective view illustrating a condition in the process of attaching the toner pack **100** to the apparatus-side shutter **109**. FIG. **21**A shows only the nozzle **102**; the pouch **101** and the pack-side shutter **103** of the toner pack **100** are omitted. FIG. **21**B is a cross-sectional view of section **16**B-**16**B of FIG. **16**A illustrating a condition when the attachment of the toner pack **100** to the mounting portion **106** is completed. For better visibility, the cut sections of the pack-side shutter **103** and cover **110** are shown with hatching in FIG. **18**A to FIG. **20**B, and the cut sections of the nozzle **102** are shown with hatching in FIG. **21**B.

(110) To attach the toner pack **100**, the user moves the toner pack **100** toward the mounting portion **106** in the mounting direction M, with the apparatus-side shutter **109** being in the covering position, and with the pack-side shutter **103** in the covering position, as shown in FIG. **17**A and

FIG. **17**B. In doing this, the user aligns the recessed portion **102***e* of the nozzle **102**, the opening **103***a* of the pack-side shutter **103**, and the positioning portion **107***a* of the first frame **107**. At the same time, the user aligns the drive transmitted portion **103***e* of the pack-side shutter **103** and the drive-transmitting portion **108***a* of the operation lever **108**.
(111) After aligning the toner pack **100** and mounting portion **106** in this way, the user moves the toner pack **100** in the mounting direction M toward the mounting portion **106**. Shortly, a small-diameter portion **109***d***2** of the center boss **109***d* of the apparatus-side shutter **109** fits with the inner circumferential surface **102***b***1** of the protruded portion **102***b* of the nozzle **102** as shown in FIG. **18**A. This determines the position of the nozzle **102** in the radial direction r relative to the apparatus-side shutter **109**.
(112) At this time, the drive-transmitting portion **108***a* of the operation lever **108** engages with the drive transmitted portion **103***e* of the pack-side shutter **103** as shown in FIG. **19**A. At the same time, a side face **110***f* and a side face **110***g* of the cover **110** come to close proximity to or contact with the surface **102***e***1** and the surface **102***e***2** that form the recessed portion **102***e* of the nozzle **102** as shown in FIG. **19**B. The drive transmitted portion **103***e* of the pack-side shutter **103** engages with the drive transmitted portion **109***e* of the apparatus-side shutter **109** and the drive-transmitting portion **108***a* of the operation lever **108** as shown in FIG. **19**A and FIG. **19**B. This makes the rotation axis A of the pack-side shutter **103** and the rotation axis B of the apparatus-side shutter **109** substantially coaxial.
(113) Since the surfaces **102***e***1** and **102***e***2** of the recessed portion **102***e* of the nozzle **102** engage respectively with the side faces **110***f* and **110***g* of the cover **110**, the nozzle **102** of the toner pack **100** does not rotate relative to the body base portion **2** including the cover **110**. In other words, the recessed portion **102***e* restricts the rotation of the nozzle **102** relative to the image forming apparatus **1** by engaging with the cover **110** of the image forming apparatus **1**, when the toner pack **100** is attached to the image forming apparatus **1**. Thus the operation lever **108**, pack-side shutter **103**, and apparatus-side shutter **109** are able to rotate substantially integrally about the rotation axis B relative to the body base portion **2** and the nozzle **102**.
(114) Specifically, when the operation lever **108** is rotated, the drive-transmitting portion **108***a* of the operation lever **108** presses the surface **103***b***1** or **103***b***2** of the pack-side shutter **103**, thereby rotating the pack-side shutter **103**. After that, the surface **103***b***1** or **103***b***2** that forms the drive transmitted portion **103***e* of the pack-side shutter **103** presses the drive transmitted portion **109***e* of the apparatus-side shutter **109**, thereby rotating the apparatus-side shutter **109**.
(115) When the attachment of the toner pack **100** to the mounting portion **106** is completed, the three radial positioning portions **103***f* (see FIG. **11**A and FIG. **13**) of the pack-side shutter **103** are in contact with the inner circumferential surface **109***h* of the apparatus-side shutter **109** (see FIG. **6**A). This determines the position of the toner pack **100** in the radial direction r on the upstream side in the mounting direction M.
(116) The protruded end face **102***b***2** of the protruded portion **102***b* of the nozzle **102** coming to abut on the pack contact surface **109***g* of the apparatus-side shutter **109** as shown in FIG. **20**A determines the position of the toner pack **100** in the mounting direction M. Alternatively, the positioning of the protruded portion **102***b* of the nozzle **102** may be achieved by a configuration in which an outer circumferential surface of the protruded portion **102***b* fits with the cylindrical portion **110***j* of the cover **110** (see FIG. **6**A and FIG. **6**B).
(117) The positioned portion **102***d* of the nozzle **102** engages with the positioning portion **107***a* of the first frame **107** as shown in FIG. **20**B. This restricts the rotation of the nozzle **102** of the toner pack **100** relative to the first frame **107** (body base portion **2**).
(118) As described in the foregoing, when the toner pack **100** is attached to the mounting portion **106**, as shown in FIG. **21**A and FIG. **21**B, the pawl **102***f* of the nozzle **102** moves from the restricting position to the non-restricting position (position shown in FIG. **21**B). More specifically, with the unlocking inclined surface **102***f***1** being pressed by the unlocking rib **109***j*, the pawl **102***f*

moves inward in the radial direction r from the restricting position toward the non-restricting position. This unlocks the rotation restriction of the pack-side shutter **103** in the direction of arrow K.
(119) [Operation of Operation Lever]
(120) FIG. **22**A is a perspective view illustrating the operation lever **108** in the closed position and the toner pack **100**. FIG. **22**B is a perspective view illustrating the operation lever **108** in the open position and the toner pack **100**. FIG. **23**A is a cross-sectional view of the toner pack **100** and mounting portion **106** when the apparatus-side shutter **109** and pack-side shutter **103** are both in the covering position. FIG. **23**B is a cross-sectional view of the toner pack **100** and mounting portion **106** when the apparatus-side shutter **109** and pack-side shutter **103** are both in the open position.
(121) As described in the foregoing, with the toner pack **100** attached to the mounting portion **106**, the operation lever **108**, pack-side shutter **103**, and apparatus-side shutter **109** are able to rotate integrally about the rotation axis B relative to the body base portion **2** and the nozzle **102**. With the toner pack **100** attached to the mounting portion **106**, and with the operation lever **108** in the closed position, the discharge port **102***a* is covered by the pack-side shutter **103**, pack-side seal **105**, and apparatus-side shutter **109**, as shown in FIG. **23**A. This configuration prevents the toner inside the pouch **101** from reaching the apparatus-side opening **117***a* of the second frame **117**.
(122) With the toner pack **100** attached to the mounting portion **106**, when the operation lever **108** is rotated in the direction of arrow Q from the closed position to the open position as shown in FIG. **22**A and FIG. **22**B, the pack-side shutter **103** and apparatus-side shutter **109** are rotated from the covering position to the open position.
(123) More specifically, the drive-transmitting portion **108***a* of the operation lever **108** presses the surface **103***b***1** of the pack-side shutter **103**. This rotates the pack-side shutter **103** with the operation lever **108** from the covering position to the open position. In other words, the pack-side shutter **103** rotates from the covering position to the open position with the rotation of the operation lever **108** by the engagement between the drive-transmitting portion **108***a* and the surface **103***b***1**. The surface **103***b***2** of the pack-side shutter **103** being rotated from the covering position to the open position presses the drive transmitted portion **109***e* of the apparatus-side shutter **109**. This rotates the apparatus-side shutter **109** with the pack-side shutter **103** from the covering position to the open position. In other words, the apparatus-side shutter **109** rotates integrally with the pack-side shutter **103** with the rotation of the operation lever **108** by the engagement between the surface **103***b***2** and the drive transmitted portion **109***e*.
(124) The pack-side shutter **103**, pack-side seal **105**, and apparatus-side shutter **109** moving together opens the discharge port **102***a* of the nozzle **102** as shown in FIG. **23**B. Namely, the pouch **101** of the toner pack **100** is communicated with the container portion **36** through the discharge port **102***a*, inlet port **109***a*, and apparatus-side opening **117***a*. As the user compresses the pouch **101**, the toner inside the pouch **101** is supplied into the container portion **36** of the developer container **32** with air through the discharge port **102***a*, inlet port **109***a*, and apparatus-side opening **117***a*.
(125) When the toner supply from the toner pack **100** to the developer container **32** is complete, the user rotates the operation lever **108** from the open position to the closed position. When the operation lever **108** is rotated from the open position to the closed position, the drive-transmitting portion **108***a* of the operation lever **108** presses the surface **103***b***2** of the pack-side shutter **103**. This rotates the pack-side shutter **103** with the operation lever **108** from the open position to the covering position. The surface **103***b***1** of the pack-side shutter **103** being rotated from the open position to the covering position presses the drive transmitted portion **109***e* of the apparatus-side shutter **109**. This rotates the apparatus-side shutter **109** with the pack-side shutter **103** from the open position to the covering position. In this way, when the operation lever **108** is rotated from the open position to the closed position, the surface **103***b***2** configures an engaging portion, and the surface **103***b***1** configures a second engaging portion.
(126) In this state, the user pulls out the toner pack **100** from the mounting portion **106** to complete

the toner supply operation.
(127) [Toner Pack Detail]
(128) A manufacturing method for the toner pack **100** according to this embodiment, and the configuration details of the joint between the nozzle body and the connector member are described with reference to FIG. **24**A to FIG. **34**B.
(129) [Toner Pack Manufacturing Process]
(130) First, a manufacturing process of the toner pack according to this embodiment is described with reference to FIG. **27**A to FIG. **28**B.
(131) FIG. **27**A is a perspective view of a pouch and a connector member separate from each other before the connector member is joined to the pouch. FIG. **27**B is a perspective view after the connector member has been joined to the pouch. FIG. **27**C is a perspective view illustrating a state of the pouch ready to be filled with the toner, with the connector member joined thereto. FIG. **28**A is an enlarged perspective view of the pouch and nozzle in a state in which the nozzle body is in a fully inserted position relative to the connector member. FIG. **28**B is an enlarged perspective view of the pouch and nozzle assembly in a state in which the nozzle body is rotated from the fully inserted position shown in FIG. **28**A to a fully engaged position.
(132) As described above, the toner pack **100** according to this embodiment includes the pouch **101** as a container member for containing toner, the nozzle **102** joined to the pouch **101**, and the pack-side shutter **103**. In this embodiment, the nozzle **102** is composed of the nozzle body **121** and the connector member **122**. In the manufacture of the toner pack **100**, the components mentioned above are assembled, as well as the pouch **101** is filled with the toner.
(133) Specifically, the connector member **122** is first attached to the pouch **101** (first assembling step) as shown in FIG. **27**A and FIG. **27**B. The attachment method is as described above. Next, the pouch **101**, in the state ready to be filled with the toner with the connector member **122** alone being joined to the pouch **101** as shown in FIG. **27**B, is filled with the toner through the opening in the connector member **122** (filling step). The nozzle body **121** is then engaged with the connector member **122** to join the nozzle **102** to the pouch **101** (form a pouch and nozzle assembly) as shown in FIG. **27**A to FIG. **28**B (second assembling step).
(134) More specifically, in assembling the nozzle body **121** to the connector member **122**, an insertion portion **121***a* is first inserted into the through hole **122***b* in the insertion direction I as shown in FIG. **27**A, and the nozzle body **121** is moved in the insertion direction I relative to the connector member **122**. The insertion direction I is parallel to the mounting direction M and opposite from the mounting direction M (parallel to and the same as the removal direction U). As shown in FIG. **28**A, the nozzle body **121** is inserted to the connector member **122** as far as to a fully inserted position where a flange **121***c* abuts on an opposite portion **122***c* as an abutted portion of the connector member **122** (insertion step). As shown in FIG. **28**B, with the flange **121***c* being maintained in contact with the opposite portion **122***c*, the nozzle body **121** is rotated relative to the connector member **122** in a rotating direction S around a rotation axis C that is parallel to the insertion direction I (rotation step). The rotation axis C is substantially the same as the rotation axes A and B.
(135) Lastly, the pack-side shutter **103** is attached. Thus the toner pack **100** shown in FIG. **9**, FIG. **10** and other drawings is complete (third assembling step).
(136) [Toner Filling]
(137) As described above, in this embodiment, the pouch **101** is filled with the toner in the state in which the connector member **122** is attached to the pouch **101** before the nozzle body **121** is attached (FIG. **27**B). Namely, the toner filling step is performed after the first assembling step and before the second assembling step.
(138) The toner filling step could be performed before the attachment of the connector member **122**, for example, through the opening portion **101***b* in the pouch **101**. However, the pouch **101** is a flexible member, and some means would be required to maintain the shape of the opening portion

101*b* during the toner filling. According to this embodiment, the connector member **122**, which is a resin component, is attached, and the opening in the connector member **122** is used as the toner fill port. Therefore, no means for maintaining the shape of the fill port is necessary.
(139) Another possibility would be to attach the nozzle to the pouch and to fill the pouch with the toner using the discharge port of the nozzle, for example. However, it may not be easy to fill the toner in some cases depending on the shape of the discharge port, and the filling operation may not be performed efficiently. In particular, with a configuration in which the discharge port **102***a* opens on one side relative to the longitudinal direction of the toner pack **100**, as in the nozzle **102** of this embodiment, efficient toner filling through the discharge port **102***a* may be difficult. This embodiment adopts a configuration in which the opening in the connector member **122**, which has a wider opening area than the discharge port **102***a*, and opens straight to the container portion **101***a* of the pouch **101** along the longitudinal direction of the toner pack **100**, is used for the toner filling. This enables easy toner filling and allows for efficient filling operation.
(140) [Engagement Structure between Nozzle Body and Connecting Member]
(141) The engagement structure between the nozzle body **121** and the connector member **122** of the toner pack **100** according to the embodiment is described with reference to FIG. **24**A to FIG. **34**B.
(142) FIG. **24**A is a perspective view of the nozzle body of the toner pack according to the embodiment. FIG. **24**B is a perspective view of the nozzle body of the toner pack according to the embodiment from a different point of view than in FIG. **24**A. FIG. **25**A is a view (plan view) of the nozzle body of the toner pack according to the embodiment seen from the opposite side from the side on which the nozzle body is inserted to and removed from the connector member. FIG. **25**B is a view (bottom view) of the nozzle body of the toner pack according to the embodiment seen from the side on which the nozzle body is inserted to and removed from the connector member. FIG. **26**A is a perspective view of the connector member of the toner pack according to the embodiment. FIG. **26**B is a perspective view of the connector member of the toner pack according to the embodiment from a different point of view than in FIG. **26**A. FIG. **29**A is a side view illustrating one side of the nozzle body where the discharge port is provided. FIG. **29**B is a cross-sectional view of the connector member including the rotation axis of relative rotation between the nozzle body and connector member. FIG. **29**C is a side view illustrating the opposite side from the side shown in FIG. **29**A of the nozzle body. FIG. **29**D is a cross-sectional view of the connector member including the rotation axis of relative rotation between the nozzle body and connector member.
(143) As shown in FIG. **26**A, FIG. **26**B, FIG. **27**A, FIG. **27**B, FIG. **29**B, and FIG. **29**D, the connector member **122** is a substantially annular member attached along the inner periphery of the opening portion **101***b* of the pouch **101**. The connector member **122** includes an insertion portion **122***a* having a shape conforming to the shape of the opening portion **101***b* of the pouch **101**, and a through hole **122***b* for communicating the container portion **101***a* of the pouch **101** with the outside when the connector member **122** is joined to the opening portion **101***b* of the pouch **101**. The connector member **122** includes the opposite portion **122***c* that is exposed to the outside of the pouch **101** in the mounting direction M when the connector member **122** is joined to the opening portion **101***b* of the pouch **101**, and that comes opposite the nozzle body **121** in the mounting direction M when the nozzle body **121** is joined to the connector member **122**.
(144) The opening portion **101***b* of the pouch **101** opens toward the mounting direction M, and with the connector member **122** joined to the opening portion **101***b*, the through hole **122***b* extends through the connector member **122** along the mounting direction M. This through hole **122***b* serves as the fill port to allow the container portion **101***a* of the pouch **101** to be filled with the toner in the toner filling step described above. The through hole **122***b* is formed by an inner circumferential surface **122***d* of the connector member **122** having a center axis parallel to the mounting direction M, and an engaged groove **124** to be described later. These constituent parts form an inserted portion **122***e* to which the insertion portion **121***a* of the nozzle body **121** is inserted during the assembly of the nozzle body **121** (second assembling step).

(145) As shown in FIG. **24**A, FIG. **24**B, FIG. **25**A, FIG. **25**B, FIG. **29**A, and FIG. **29**C, the nozzle body **121** includes the insertion portion **121***a*, a mounted portion **121***b*, and the flange **121***c*. The insertion portion **121***a* is a part of the nozzle body **121** that is inserted into the through hole **122***b* of the connector member **122**. The mounted portion **121***b* is a part formed on the opposite side of the flange **121***c* from the insertion portion **121***a*, which is a part of the nozzle body **121** attached to the mounting portion **106** of the image forming apparatus **1** when the toner pack **100** is attached to the image forming apparatus **1**. The mounted portion **121***b* is a part exposed to the outside of the pouch **101** when the nozzle body **121** is joined to the connector member **122** during the assembly of the toner pack **100**, where the discharge port **102***a* and the protruded portion **102***b* are provided.
(146) The flange **121***c* is provided between the insertion portion **121***a* and the mounted portion **121***b*, i.e., upstream of the insertion portion **121***a* in the insertion direction I in which the insertion portion **121***a* is inserted into the through hole **122***b*, and extends in a direction perpendicular to the insertion direction I. The flange **121***c* has, as an abutting portion, a surface facing the opposite portion **122***c* of the connector member **122** in the insertion direction I when the insertion portion **121***a* is inserted into the through hole **122***b*. When the insertion portion **121***a* is inserted into the through hole **122***b*, the flange **121***c* abuts on the opposite portion **122***c* in the insertion direction I, thereby defining the fully inserted position of the nozzle body **121** relative to the connector member **122**.
(147) FIG. **34**A is a cross-sectional perspective view of the pouch and nozzle assembly in the vicinity of the nozzle, showing section **34**A-**34**A of FIG. **34**B. FIG. **34**B is a cross-sectional view of the configuration of a seal member, illustrating section **34**B-**34**B of FIG. **34**A. As shown in FIG. **34**B, an annular seal member **127** is arranged between the flange **121***c* of the nozzle body **121** and the opposite portion **122***c* of the connector member **122** such as to surround the outer periphery of the through hole **122***b*. The seal member **127** is compressed between the flange **121***c* and the opposite portion **122***c* in the insertion direction I, thereby sealing the gap between the flange **121***c* and the opposite portion **122***c* and preventing toner leakage.
(148) The engagement structure between the nozzle body **121** and the connector member **122** is described in more detail with reference to FIG. **24**A to FIG. **26**B and FIG. **29**A to FIG. **33**B.
(149) FIG. **30**A is an enlarged perspective view illustrating a configuration of an inner engaging protrusion. FIG. **30**B is an enlarged perspective view illustrating a configuration of an engaged groove. FIG. **31**A is a schematic enlarged perspective view illustrating the inside of the opening in the connector member that is attached to the pouch, with only the inner engaging protrusion of the nozzle body being shown, when the nozzle body is in a fully inserted position relative to the connector member. FIG. **31**B is a cross-sectional view of section **31**B-**31**B of FIG. **31**A. FIG. **31**C is a schematic enlarged perspective view illustrating the inside of the opening in the connector member that is attached to the pouch, with only the inner engaging protrusion of the nozzle body being shown, when the nozzle body is in a fully engaged position relative to the connector member. FIG. **31**D is a cross-sectional view of section **31**D-**31**D of FIG. **31**C. FIG. **32**A is an enlarged side view of the pouch and nozzle illustrating a state of a second engagement structure (an outer engaging protrusion of the nozzle body and an engaged protrusion of the connector member) when the nozzle body is in the fully inserted position. FIG. **32**B is a view (bottom view) of the pouch and nozzle shown in FIG. **32**A in the insertion direction. FIG. **32**C is an enlarged side view of the pouch and nozzle assembly illustrating a state of the second engagement structure (outer engaging protrusion of the nozzle body and engaged protrusion of the connector member) when the nozzle body is in the fully engaged position. FIG. **32**D is a view (bottom view) of the pouch and nozzle assembly shown in FIG. **32**C in the insertion direction. FIG. **33**A is an enlarged perspective view illustrating a state of the second engagement structure (outer engaging protrusion of the nozzle body and engaged protrusion of the connector member) when the nozzle body is in the fully engaged position. FIG. **33**B is an enlarged side view illustrating a state of the second engagement structure (outer engaging protrusion of the nozzle body and engaged protrusion of the connector

member) when the nozzle body is in the fully engaged position.
(150) As described above, when assembling the nozzle body **121** to the connector member **122**, the nozzle body **121** is inserted to the connector member **122** as far as to the fully inserted position, after which the nozzle body **121** is rotated relative to the connector member **122** in the rotating direction S around the rotation axis C. Between the nozzle body **121** and the connector member **122** is provided an engagement structure designed to come into engagement with each other by this rotation, i.e., the nozzle body **121** is joined to the connector member **122** by this engagement structure. Between the nozzle body **121** and the connector member **122** are provided at least two types of engagement structure at different radial distances from the rotation axis C, as the engagement structure for forming this joint.
(151) As shown in FIG. **24**A to FIG. **26**B and FIG. **29**A to FIG. **29**D, the nozzle body **121** includes the inner engaging protrusion **123** as a first engaging portion provided on a radially inner side, and an outer engaging protrusion **125** as a second engaging portion provided on a radially outer side, in the radial direction with respect to the rotation axis C. Corresponding to these parts, the connector member **122** includes the engaged groove **124** as a first engaged portion, and an engaged protrusion **126** as a second engaged portion. The inner engaging protrusion **123** engages with the engaged groove **124**, and the outer engaging protrusion **125** engages with the engaged protrusion **126**. Namely, the inner engaging protrusion **123** and the engaged groove **124** form a radially inner side engagement structure (first engagement structure), and the outer engaging protrusion **125** and the engaged protrusion **126** form a radially outer side engagement structure (second engagement structure). Both engagement structures are designed to come to a mutually engaged state after going through a process of elastic deformation during the relative rotation between the nozzle body **121** and the connector member **122** about the predetermined rotation axis C as the center axis. In the engaged state, the nozzle body **121** and the connector member **122** are restricted from moving relative to each other in the direction of the rotation axis.
(152) [First Engagement Structure]
(153) The engagement structure (first engagement structure) formed by the inner engaging protrusion **123** and the engaged groove **124** is configured to keep them permanently fixed together. After the nozzle body **121** has reached the fully inserted position relative to the connector member **122**, as they are rotated relative to each other, the inner engaging protrusion **123** and the engaged groove **124** can take various relative rotation phases including a disengagement phase, deformation phase, and engagement phase.
(154) In the state in which the nozzle body **121** has reached the fully inserted position, the inner engaging protrusion **123** and the engaged groove **124** are in the disengagement phase. When the nozzle body **121** is rotated relative to the connector member **122** in the rotating direction S, the inner engaging protrusion **123** and the engaged groove **124** transition from the disengagement phase to the engagement phase, via the deformation phase. The inner engaging protrusion **123** and the engaged groove **124** are configured to reach the engagement phase by some elastic deformation occurring at least in one of the inner engaging protrusion **123** and the engaged groove **124** in the deformation phase.
(155) In the state in which the inner engaging protrusion **123** and the engaged groove **124** are in the engagement phase, the nozzle body **121** is restricted from moving relative to the connector member **122** in the circumferential direction and in the axial direction of the rotation center. Once having reached the engagement phase, the inner engaging protrusion **123** and the engaged groove **124** never return to the deformation phase or disengagement phase even when the nozzle body **121** is rotated in the opposite direction (second direction) from the rotating direction S (first direction) relative to the connector member **122**.
(156) The inner engaging protrusion **123** protrudes radially from an outer circumferential surface of the insertion portion **121***a* of the nozzle body **121**, as shown in FIG. **24**A to FIG. **25**B. On the outer circumferential surface of the insertion portion **121***a*, the inner engaging protrusions **123** extend in

a spiral line so that they extend downstream in the insertion direction I in which the insertion portion **121***a* is inserted into the through hole **122***b* as well as downstream in the rotating direction S in which the nozzle body **121** is rotated when the nozzle body **121** is joined to the connector member **122**.
(157) The inner engaging protrusion **123** includes a first side face **123***d* and a second side face **123***e* at the leading end and trailing end of the rotating direction S, respectively, and a third side face **123***f* at the rear end in the insertion direction I, as shown in FIG. **29**A, FIG. **29**C, and FIG. **30**A. The first side face **123***d* and the second side face **123***e* are side faces extending in the insertion direction I. The third side face **123***f* is a side face inclined to both of the insertion direction I and rotating direction S; it extends downstream in the insertion direction I as well as downstream in the rotating direction S.
(158) Between the first side face **123***c* and the third side face **123***f*, the inner engaging protrusion **123** further includes a force-receiving surface **123***b* (first guided inclined surface) that is a side face inclined to both of the first and third side faces. The force-receiving surface **123***b* is a surface for receiving a pressing force from the engaged groove **124**. This pressing force can cause a deformation in at least one of the inner engaging protrusion **123** and the engaged groove **124** in the deformation phase for allowing a relative movement of the inner engaging protrusion **123** and the engaged groove **124** to reach the engagement phase.
(159) The engaged groove **124** is a groove provided inside the through hole **122***b* of the connector member **122**, i.e., in the inner circumferential surface **122***d* that forms the through hole **122***b*, as shown in FIG. **26**A and FIG. **26**B. The engaged groove **124** is recessed radially outward from the inner circumferential surface **122***d* in the radial direction with respect to the center axis of the inner circumferential surface **122***d*, which is parallel to the insertion direction I.
(160) The engaged groove **124** is configured to generally include an insertion guide portion **124***a*, a deformation guide portion **124***b*, and an engagement retaining portion **124***c* as shown in FIG. **29**B, FIG. **29**D, and FIG. **30**B. As shown in FIG. **31**A to FIG. **31**D, the inner engaging protrusion **123** led into the engaged groove **124** is guided from the insertion guide portion **124***a* toward the engagement retaining portion **124***c* by the relative rotation of the nozzle body **121** relative to the connector member **122** in the rotating direction S, via a contact with a force-applying surface of the deformation guide portion **124***b*.
(161) Namely, when the insertion portion **121***a* is inserted into the through hole **122***b* in the insertion direction I, and the nozzle body **121** is moved in the insertion direction I relative to the connector member **122**, the inner engaging protrusion **123** and engaged groove **124** come to a relative position (first position) shown in FIG. **31**A and FIG. **31**B. In this position, the inner engaging protrusion **123** and the engaged groove **124** are fitted with each other such as to allow relative movement of the nozzle body **121** and the connector member (coupling member) **122** in a direction along the mounting direction M (third direction, predetermined direction). This direction (third direction, predetermined direction) is also a direction in which the pouch **101** (container portion **101***a*) of the toner pack **100** and the nozzle **102** (nozzle body **121** and inner engaging protrusion **123**) align. By moving the nozzle body **121** relative to the connector member **122** in the rotating direction S (first direction in the circumferential direction, which is a fourth direction intersecting the third direction, or intersecting direction) from this relative position (first position), the inner engaging protrusion **123** and engaged groove **124** come to a relative position (second position) shown in FIG. **31**C and FIG. **31**D. In this position, the inner engaging protrusion **123** and the engaged groove **124** are fitted with each other such as to restrict relative movement of the nozzle body **121** and connector member **122**. Specifically, a third groove side face **124***f* restricts the relative movement of the nozzle body **121** and the connector member **122** in a direction along the mounting direction M (third direction, predetermined direction). A second groove side face **124***e* restricts the relative movement of the nozzle body **121** and the connector member **122** in the rotating direction S (second direction in the circumferential direction, which is the fourth direction

or intersecting direction).

(162) The insertion guide portion **124***a* includes an insertion hole open to a direction parallel to the insertion direction I in the end face of the opposite portion **122***c* on the opposite side from the insertion direction I of the connector member **122**. The insertion guide portion **124***a* extends from this insertion hole in the insertion direction I, to draw in and guide the inner engaging protrusion **123** in the insertion direction I, when the insertion portion **121***a* of the nozzle body **121** is inserted into the through hole **122***b*.

(163) The engaged groove **124** has a portion extending in the rotating direction S beyond the insertion guide portion **124***a* (downstream side in the insertion direction I), with the deformation guide portion **124***b* right before the engagement retaining portion **124***c*. The deformation guide portion **124***b* is configured to protrude downstream in the insertion direction I into the engagement retaining portion **124***c*, and includes a first inclined surface **124***b***1** and a second inclined surface **124***b***2**, which form part of the groove side face of the engaged groove **124**, as the force-applying surface (first guide inclined surface).

(164) The first inclined surface **124***b***1** and second inclined surface **124***b***2** extending in the direction in which they guide the inner engaging protrusion **123** have respective inclination angles that are reduced stepwise relative to the circumferential direction. The first inclined surface **124***b***1** and second inclined surface **124***b***2** are groove side faces inclined to both of the insertion direction I and rotating direction S, extending downstream in the insertion direction I as well as downstream in the rotating direction S. The first inclined surface **124***b***1** and the second inclined surface **124***b***2** face the force-receiving surface **123***b* of the inner engaging protrusion **123** in the deformation phase from the opposite directions from the insertion direction I and rotating direction S, respectively. The inner engaging protrusion **123** is first guided by the first inclined surface **124***b***1**, and then by the second inclined surface **124***b***2**. The second inclined surface **124***b***2** has a smaller inclination angle than the first inclined surface **124***b***1** relative to the circumferential direction.

(165) When the inner engaging protrusion **123** moves in the rotating direction S with the relative rotation between the nozzle body **121** and the connector member **122**, the force-receiving surface **123***b* comes into contact with the first inclined surface **124***b***1**. The pressing force generated between the force-receiving surface **123***b* and the first inclined surface **124***b***1** brings about an elastic deformation at least in one of the inner engaging protrusion **123** and the engaged groove **124**. This deformation causes the inner engaging protrusion **123** to shift downstream in the insertion direction I relative to the deformation guide portion **124***b* of the engaged groove **124**.

(166) As the relative rotation between the nozzle body **121** and the connector member **122** progresses, with the force-receiving surface **123***b* and the first inclined surface **124***b***1** sliding against each other, the inner engaging protrusion **123** rides onto the deformation guide portion **124***b* downstream in the insertion direction I. Namely, the third side face **123***f* of the inner engaging protrusion **123** comes into contact with and slides against the second inclined surface **124***b***2** of the deformation guide portion **124***b* of the engaged groove **124**. The deformation of at least one of the inner engaging protrusion **123** and the engaged groove **124**, caused by the pressing force generated between the force-receiving surface **123***b* and the second inclined surface **124***b***2**, causes the inner engaging protrusion **123** to shift further downstream in the insertion direction I relative to the deformation guide portion **124***b* of the engaged groove **124**.

(167) As the relative rotation between the nozzle body **121** and the connector member **122** progresses from this state, the inner engaging protrusion **123** moves further downstream in the rotating direction S beyond the deformation guide portion **124***b*, whereupon the pressing force is released. The inner engaging protrusion **123**, freed from the deformed state, shifts in the opposite direction from the insertion direction I relative to the engaged groove **124**. Namely, the inner engaging protrusion **123** moves in the rotating direction S relative to the engaged groove **124** by the deformation of at least one of the inner engaging protrusion **123** and the engaged groove **124** such as to ride over the deformation guide portion **124***b*, and fits into the engagement retaining

portion **124***c*. The relative rotation phase of the inner engaging protrusion **123** and the engaged groove **124** thus transitions from the deformation phase to the engagement phase.
(168) The engagement retaining portion **124***c* is formed to increase in width, which is the groove width of the engaged groove **124** in the insertion direction I, in the opposite direction from the insertion direction I relative to the deformation guide portion **124***b*. The elastic deformation of the inner engaging protrusion **123** or engaged groove **124** that occurred in the deformation phase is released by the transition of their relative rotation phase to the engagement phase, whereby the inner engaging protrusion **123** comes to fit into the engagement retaining portion **124***c*.
(169) The engagement retaining portion **124***c* includes a first groove side face **124***d* and a second groove side face **124***e* arranged along the circumferential direction of the inner circumferential surface **122***d*. The first groove side face **124***d* and the second groove side face **124***e* face the inner engaging protrusion **123** in the engagement phase in the opposite direction (first direction) from the rotating direction S, and in the rotating direction S (second direction), respectively. The first groove side face **124***d* is a groove side face extending along the insertion direction I, and faces the first side face **123***d* of the inner engaging protrusion **123** in the opposite direction from the rotating direction S, serving as a first circumferential restricting portion. The second groove side face **124***e* is also a groove side face extending along the insertion direction I, and faces the second side face **123***e* of the inner engaging protrusion **123** in the rotating direction S, serving as a second circumferential restricting portion (second restricting portion).
(170) These surfaces facing each other in the circumferential direction between the inner engaging protrusion **123** and the engagement retaining portion **124***c* restrict relative rotational movement in the circumferential direction of the inner engaging protrusion **123** and engaged groove **124** in the engagement phase. While the circumferentially opposite surfaces between the inner engaging protrusion **123** and the engagement retaining portion **124***c* should ideally face each other without a gap, they may be configured to face each other with a slight gap. Namely, these parts may be configured to loosely fit to each other as long as the range of relative rotation allowed to the nozzle body **121** and connector member **122** after they have engaged does not adversely affect the function of the toner pack **100**.
(171) The engagement retaining portion **124***c* has a third groove side face **124***f* that faces the inner engaging protrusion **123** in the engagement phase in the insertion direction I. The third groove side face **124***f* is a groove side face inclined to both of the insertion direction I and rotating direction S, conforming to the third side face **123***f* of the inner engaging protrusion **123**, i.e., the third groove side face extends downstream in the insertion direction I as well as downstream in the rotating direction S.
(172) The third groove side face **124***f*, serving as a counter-insertion direction restricting portion (first restricting portion), abuts on the third side face **123***f* of the inner engaging protrusion **123** in the engagement phase in the insertion direction I. This restricts the relative movement of the nozzle body **121** in the counter-insertion direction, which is the opposite direction from the insertion direction I, away from the connector member **122**. Meanwhile, the relative movement of the nozzle body **121** relative to the connector member **122** in the insertion direction I is restricted, as described above, by the opposite portion **122***c* of the connector member **122** serving as an insertion direction restricting portion, abutting on the flange **121***c* of the nozzle body **121**, in the counter-insertion direction. Thus the movement of the nozzle body **121** relative to the connector member **122** is restricted in both the insertion I and the counter-insertion direction.
(173) The third groove side face **124***f* is inclined such as to press the inner engaging protrusion **123** downstream in the insertion direction I as well as downstream in the rotating direction S. The wedge effect of this inclined configuration further ensures the restriction of the relative movement between the inner engaging protrusion **123** and the engaged groove **124**, and enhances the air tightness between the nozzle body **121** and connector member **122** in the insertion direction I. Namely, any dimensional errors can be absorbed so that a reliable tight contact can be achieved

between the nozzle body **121** and connector member **122**.
(174) The inner engaging protrusion **123** is provided in plural different positions in the circumferential direction on the outer circumferential surface of the insertion portion **121***a* (at different phase positions about the center axis of the insertion portion **121***a*). In this embodiment, the inner engaging protrusion is provided at four circumferentially equidistant positions. Likewise, the engaged groove **124** is provided at four circumferentially equidistant positions corresponding to the inner engaging protrusions (first engaging protrusions) **123**. The number of engagement structures composed of the inner engaging protrusion **123** and engaged groove **124** is not limited to four; it may be three or less, or five or more.
(175) In this embodiment, the position in the insertion direction I of one of the four sets of engagement structure is made different than those of the other sets, as a configuration for preventing incorrect attachment of the nozzle body **121** to the connector member **122**. Namely, one (**123**B) of the four inner engaging protrusions **123** in the insertion direction is displaced downstream in the insertion direction I relative to the other three inner engaging protrusions **123**A. Accordingly, the position in the insertion direction I of the engagement retaining portion **124***c* in the engaged groove **124**B corresponding to the inner engaging protrusion **123**B is displaced downstream in the insertion direction I relative to the positions of the engagement retaining portions **124***c* in the other three engaged grooves **124**A. Therefore, the nozzle body **121** cannot be assembled to the connector member **122** unless the nozzle body **121** and connector member **122** are positioned in a relative phase around the insertion direction I where the inner engaging protrusion **123**B and the engaged groove **124**B are aligned. The phase (orientation) around the insertion direction I of the nozzle body **121** relative to the connector member **122** when the nozzle body is engaged with the connector member **122**, i.e., the phase around the insertion direction I of the nozzle body **121** relative to the pouch **101**, is thus made constant (is determined to one orientation).
(176) The configuration for preventing incorrect attachment is not limited to the one described above. For example, for displacing the inner engaging protrusions **123**A and inner engaging protrusion **123**B in the insertion direction I, the inner engaging protrusions **123**A and inner engaging protrusion **123**B may be arranged such that the inner engaging protrusion **123**B is displaced upstream in the insertion direction I relative to the inner engaging protrusions **123**A. Alternatively, two adjacent inner engaging protrusions **123** may be positioned in the insertion direction I differently from the other two inner engaging protrusions, for example. Three or less, or five or more inner engaging protrusions **123** may be provided. In this case, the inner engaging protrusions **123**A and inner engaging protrusion **123**B may be combined in any way as long as the function of preventing incorrect attachment is achieved. The positions of the inner engaging protrusions **123** may be displaced in the insertion direction I in three or more steps instead of two steps as in this embodiment.
(177) Instead of providing the engaging protrusions on the nozzle body **121** and forming the engaged grooves in the connector member **122** as in this embodiment, the engaging protrusions may be provided to the connector member **122**, and the engaged grooves may be formed in the nozzle body **121**. In the latter case, however, the engaged grooves formed in the nozzle body **121** will be in communication with the container portion **101***a* of the pouch **101**. Therefore, it may be necessary to consider providing a sealing structure to prevent the engagement structure from becoming a leakage path of toner.
(178) [Second Engagement Structure]
(179) The engagement structure (second engagement structure) formed by the outer engaging protrusion **125** (third protrusion) and the engaged protrusion **126** (second protrusion) is also configured to keep them permanently fixed together. After the nozzle body **121** has reached the fully inserted position relative to the connector member **122**, as they are rotated relative to each other, the outer engaging protrusion **125** and the engaged protrusion **126** can take various relative rotation phases including a disengagement phase, deformation phase, and engagement phase.

(180) In the state in which the nozzle body **121** has reached the fully inserted position, the outer engaging protrusion **125** and the engaged protrusion **126** are in the disengagement phase. When the nozzle body **121** is rotated relative to the connector member **122** in the rotating direction S, the outer engaging protrusion **125** and the engaged protrusion **126** transition from the disengagement phase to the engagement phase, via the deformation phase. The outer engaging protrusion **125** and the engaged protrusion **126** are configured to reach the engagement phase by some elastic deformation occurring at least in one of the outer engaging protrusion **125** and the engaged protrusion **126** in the deformation phase.
(181) In the state in which the outer engaging protrusion **125** and the engaged protrusion **126** are in the engagement phase, the nozzle body **121** is restricted from moving relative to the connector member **122** in the circumferential direction and in the axial direction of the rotation center. Once having reached the engagement phase, the outer engaging protrusion **125** and the engaged protrusion **126** never return to the deformation phase or disengagement phase even when the nozzle body **121** is rotated in the opposite direction from the rotating direction S relative to the connector member **122**, unless some external force that can cause deformation of the outer engaging protrusion **125** or engaged protrusion **126** is applied.
(182) The outer engaging protrusion **125** is a protrusion in a substantially delta wing form projecting radially outward from an outer peripheral edge of the flange **121***c* of the nozzle body **121**, as shown in FIG. **24**A to FIG. **25**B and FIG. **32**A to FIG. **32**D. Specifically, the outer engaging protrusion **125** protrudes radially more toward the opposite direction from the rotating direction S, which is the direction of relative rotation of the nozzle body **121** relative to the connector member **122**.
(183) The outer engaging protrusion **125** includes a force-receiving surface **125***a*, a sliding surface **125***b*, and an engaging surface **125***c* as shown in FIG. **33**A and FIG. **33**B. The force-receiving surface **125***a* (second guided inclined surface) is an end face of the outer engaging protrusion **125** at the distal end in the rotating direction S; it is an inclined surface inclined radially outward as it extends upstream in the rotating direction S, and extending along the insertion direction I. The sliding surface **125***b* is an end face of the outer engaging protrusion **125** downstream in the insertion direction I, extending in a direction perpendicular to the insertion direction I. The engaging surface **125***c* is an end face of the outer engaging protrusion **125** at the rear end in the rotating direction S, extending in the insertion direction I and in the radial direction.
(184) The engaged protrusion **126** is a substantially triangular rib-like protrusion projecting from the opposite portion **122***c* of the connector member **122** in the opposite direction from the insertion direction I, as shown in FIG. **24**A to FIG. **25**B and FIG. **32**A to FIG. **32**D. Specifically, the engaged protrusion is located radially more outside than the outer circumference of the flange **121***c* of the nozzle body **121**, and protrudes more in the counter-insertion direction opposite from the insertion direction I as it extends in the rotating direction S, which is the direction of relative rotation of the nozzle body **121** relative to the connector member **122**.
(185) The engaged protrusion **126** includes a force-applying surface **126***a*, a slid surface **126***b*, and an engaged surface **126***c* as shown in FIG. **33**A and FIG. **33**B. The force-applying surface **126***a* (second guide inclined surface) is an inclined surface of the engaged protrusion **126** upstream in the rotating direction S; it is an inclined surface inclined to protrude in the counter-insertion direction more from the opposite portion **122***c* as it extends downstream in the rotating direction S, and extending along the radial direction. The slid surface **126***b* is an end face of the engaged protrusion **126** upstream in the insertion direction I, extending in a direction perpendicular to the insertion direction I. The engaged surface **126***c* is an end face of the engaged protrusion **126** downstream in the rotating direction S, extending in the insertion direction I and in the radial direction.
(186) When the outer engaging protrusion **125** moves in the rotating direction S with the relative rotation between the nozzle body **121** and the connector member **122**, the force-receiving surface **125***a* comes into contact with the force-applying surface **126***a*. The deformation of at least one of

the outer engaging protrusion **125** and the engaged protrusion **126**, caused by the pressing force generated between the force-receiving surface **125***a* and the force-applying surface **126***a*, causes the outer engaging protrusion **125** to shift downstream in the insertion direction I relative to the engaged protrusion **126**.

(187) As the relative rotation between the nozzle body **121** and the connector member **122** progresses, with the force-receiving surface **125***a* and the force-applying surface **126***a* sliding against each other, the outer engaging protrusion **125** rides onto the engaged protrusion **126** downstream in the insertion direction I. Namely, the sliding surface **125***b* of the outer engaging protrusion **125** comes to face in the insertion direction I, makes contact with, and slides against the slid surface **126***b* of the engaged protrusion **126**.

(188) As the relative rotation between the nozzle body **121** and the connector member **122** progresses from this state, the outer engaging protrusion **125** moves further downstream in the rotating direction S beyond the engaged protrusion **126**, whereupon the pressing force is released. The outer engaging protrusion **125**, freed from the deformed state, shifts in the insertion direction I relative to the engaged protrusion **126**. Namely, the outer engaging protrusion **125** moves downstream in the rotating direction S relative to the engaged protrusion **126** by the deformation of at least one of the outer engaging protrusion **125** and the engaged protrusion **126** such as to ride over the engaged protrusion **126**. The engaging surface **125***c* of the outer engaging protrusion **125** serving as a fourth circumferential restricting portion comes to face the engaged surface **126***c* of the engaged protrusion **126** serving as a third circumferential restricting portion in the opposite direction from the rotating direction S. Thus the relative rotation of the nozzle body **121** relative to the connector member **122** is restricted in the opposite direction from the rotating direction S. Namely, the relative rotation phase of the outer engaging protrusion **125** and the engaged protrusion **126** transitions from the deformation phase to the engagement phase.

(189) The second engagement structure made up of the outer engaging protrusion **125** and engaged protrusion **126** is provided at plural different positions in the circumferential direction of the relative rotation between the nozzle body **121** and the connector member **122** about the rotation axis C (at different phases about the center axis of the insertion portion **121***a*). In this embodiment, the engagement structure is provided at two circumferential locations. The number of locations where the second engagement structure is provided is not limited to two; it may be one, or three or more.

(190) The first engagement structure described above is able to restrict the relative movement of the nozzle body **121** relative to the connector member **122** both in the insertion direction I and in the opposite or counter-insertion direction (release direction). Relative rotational movement about the rotation axis extending along the insertion direction I between the nozzle body **121** and the connector member **122** is also restricted. Restriction of relative rotation can be achieved by the second engagement structure, too, and this restriction of relative rotation consequently restricts the relative movement in insertion and release directions of the nozzle body **121** relative to the connector member **122**.

(191) The first engagement structure that restricts both the relative movement in insertion and release directions and the relative rotational movement is positioned radially inner than the second engagement structure in the radial direction with respect to the rotation axis of the relative rotation. This arrangement allows the rotational torque required for achieving the engaged state to be relatively smaller than in other arrangements (e.g., where the radial positions of the first and second engagement structures are reversed). In particular, the force required to cause deformation for allowing the inner engaging protrusion **123** to ride over the deformation guide portion **124***b* of the engaged groove **124** can be reduced, which facilitates the assembling operation.

(192) The first engagement structure made up of the inner engaging protrusion **123** and the engaged groove **124** is designed such as not to be exposed to the outside after the nozzle body **121** is assembled. That is, the first engagement structure is positioned inside the through hole **122***b* of

the inserted portion **122***e* of the connector member **122**. Moreover, the outer peripheral portion of the connector member **122** and the flange **121***c* of the nozzle body **121** function as a cover portion that covers the engagement portion between the inner engaging protrusion **123** and engaged groove **124** from outside. This prevents access from outside to the first engagement structure, so that the first engagement structure can be permanently maintained unless it is broken up. As long as access to the first engagement structure can be prevented, it does not matter whether the first engagement structure is visible from outside. Namely, even if the engagement portion of the first engagement structure is visible through a small gap, for example, such a configuration does not affect the effects of the present invention as long as the gap is of such a size that it can prevent the user from accessing the first engagement structure.
(193) In this embodiment, the first engagement structure and second engagement structure are configured such that the inner engaging protrusion **123** and the engaged groove **124** come to engage with each other at the same time as when the outer engaging protrusion **125** and the engaged protrusion **126** come to engage with each other. While the first engagement structure is not exposed to the outside, the second engagement structure is provided outside the inserted portion **122***e* of the connector member **122**, i.e., exposed to the outside. This makes it possible to check, indirectly, the engaged state of the first engagement structure that is not visible from outside, by checking the engaged state of the second engagement structure.
Variation Examples
(194) Some variation examples of the first engagement structure are described with reference to FIG. **35**A to FIG. **36**D. Features similar to those of the embodiment described above are given the same reference numerals in the variation examples.
(195) FIG. **35**A is a side view illustrating one side of the nozzle body where the discharge port is provided in a variation example. FIG. **35**B is a cross-sectional view of the connector member including the rotation axis of relative rotation between the nozzle body and connector member. FIG. **35**C is a side view illustrating the opposite side from the side shown in FIG. **35**A of the nozzle body in the variation example. FIG. **35**D is a cross-sectional view of the connector member including the rotation axis of relative rotation between the nozzle body and connector member. FIG. **36**A is a schematic enlarged perspective view illustrating the inside of the opening in the connector member that is attached to the pouch, with only the inner engaging protrusion of the nozzle body being shown, when the nozzle body is in a fully inserted position relative to the connector member, in the variation example. FIG. **36**B is a cross-sectional view of section **36**B-**36**B of FIG. **36**A. FIG. **36**C is a schematic enlarged perspective view illustrating the inside of the opening in the connector member that is attached to the pouch, with only the inner engaging protrusion of the nozzle body being shown, when the nozzle body is in a fully engaged position relative to the connector member, in the variation example. FIG. **36**D is a cross-sectional view of section **36**D-**36**D of FIG. **36**C.
(196) While the inner engaging protrusion **123** is an inclined protrusion along a spiral line in the above embodiment, the form of the inner engaging protrusion **123** is not limited to this configuration. For example, the third side face **123***f***2** may be a surface extending perpendicularly to the rotation axis of the rotating direction S, instead of an inclined surface inclined to both of the insertion direction I and rotating direction S like the third side face **123***f* in the above embodiment. Namely, the inner engaging protrusion **1232** may have a shape generally extending in the circumferential direction. In this case, the third groove side face **124***f***2** of the engagement retaining portion **124***c* of the engaged groove **1242**, which faces the third side face **123***f***2**, may be a surface extending perpendicularly to the rotation axis of the rotating direction S, corresponding to the third side face **123***f***2**.
Other Embodiments
(197) While the pouch **101** of the toner pack **100** contains toner in all of the embodiments and examples described above, this may not be necessarily so. For example, the pouch **101** may contain

other contents than toner such as ink, powder, or other fluids. The powder that can be contained in the pouch **101** is not limited to toner. In the case where the pouch **101** contains ink, the toner pack **100** may be attached to an inkjet type image forming apparatus.
(198) While the pack-side shutter **103** and the apparatus-side shutter **109** are configured to be rotatable between a covering position and an open position about the rotation axis A or B in all of the forms described above, this configuration is not limiting. For example, the pack-side shutter **103** and the apparatus-side shutter **109** can be configured to be movable between a covering position and an open position straight in parallel with the mounting direction M.
(199) While the pack-side shutter **103** is configured to open the discharge port **102***a* of the nozzle **102** only in the open position in all of the forms described above, this configuration is not limiting. For example, the pack-side shutter **103** can be a rotating member that opens the discharge port **102***a* of the nozzle **102** irrespective of its rotating position. In this case, the discharge port **102***a* of the nozzle **102** may be closed with a seal when the toner pack **100** is not attached to the mounting portion **106**, and the seal may be removed by the operation of attaching the toner pack to the mounting portion **106**, or after the attachment. Alternatively, the toner pack **100** can be configured without the pack-side shutter **103**.
(200) The present invention is not limited to the embodiments described above and can be changed and modified in various ways without departing from the spirit and scope of the present invention. Accordingly, the following claims are appended to make the scope of the present invention known.
(201) The present invention can improve the ease of assembly of developer containers while preventing developer leakage from the container.
(202) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

Claims

1. A developer container comprising: a container member including a container portion containing developer, and an opening portion that opens the container portion; a connector member attached to the opening portion; and a discharge path forming member connected to the container member via the connector member and forming a discharge path for discharging the developer from the container portion, the connector member including an inserted portion having a through hole extending therethrough, and an engaged portion, the discharge path forming member including an insertion portion inserted into the through hole of the inserted portion, and an engaging portion provided correspondingly to the engaged portion, the engaged portion and the engaging portion being configured to come to a mutually engaged state through a process in which at least one of the connector member and the discharge path forming member is elastically deformed by a relative rotation between the connector member and the discharge path forming member about a predetermined rotation axis as a center axis, in a state where the insertion portion is inserted into the inserted portion, and to restrict, in the engaged state, a relative movement of the connector member and the discharge path forming member in a direction of the rotation axis.
2. The developer container according to claim 1, wherein the engaged portion includes a first engaged portion provided inside the through hole of the inserted portion, and a second engaged portion provided outside the inserted portion, and the engaging portion includes a first engaging portion provided correspondingly to the first engaged portion, and a second engaging portion provided correspondingly to the second engaged portion.
3. The developer container according to claim 2, wherein a relative movement between the connector member and the discharge path forming member in a case of causing the engaged portion and the engaging portion to engage with each other is a relative rotation about an axial line

as the center axis, the axial line extending along an insertion direction in which the discharge path forming member is inserted into the connector member, and the engaged state is achieved by the discharge path forming member being rotated relative to the connector member in a first direction, and the engaged portion and the engaging portion in the mutually engaged state restrict a relative rotation between the connector member and the discharge path forming member in which the discharge path forming member rotates relative to the connector member in an opposite direction from the first direction.
4. The developer container according to claim 3, wherein the connector member includes an abutted portion that faces the discharge path forming member in an opposite direction from an insertion direction in which the insertion portion is inserted into the inserted portion, the discharge path forming member includes an abutting portion that faces the abutted portion in the insertion direction, the abutted portion and the abutting portion are in a state of abutting each other in the insertion direction at least in a state where the engaged portion and the engaging portion are engaged, and at least the first engaged portion and the first engaging portion, in a state of mutual engagement, restrict a relative movement of the discharge path forming member from being away from the connector member in a counter-insertion direction that is opposite from the insertion direction.
5. The developer container according to claim 4, further comprising a seal member sealing a gap between the abutted portion and the abutting portion by being compressed between the abutted portion and the abutting portion in the insertion direction.
6. The developer container according to claim 3, wherein The second engaged portion is positioned farther away from the center axis than the first engaged portion in a direction perpendicular to the center axis, and the second engaging portion is positioned farther away from the center axis than the first engaging portion in a direction perpendicular to the center axis.
7. The developer container according to claim 3, wherein the engaged portion and the engaging portion can implement a relative rotation phase during the relative rotation between the connector member and the discharge path forming member, the relative rotation phase including: a disengagement phase in which the inserting portion is in a state of being inserted in the inserted portion, and the engaged portion is not engaged with the engaging portion; a deformation phase in which the engaged portion is in a phase of not being engaged with the engaging portion, and at least one of the connector member and the discharge path forming member undergoes an elastic deformation; and an engagement phase in which the elastic deformation is released and the engaged portion engages with the engaging portion, and wherein the connector member includes a first guide inclined surface positioned upstream in the insertion direction relative to the first engaging portion and making contact with the first engaging portion in a case where the relative rotation phase is the deformation phase, the first guide inclined surface extending in a direction inclined relative to both of the insertion direction and a circumferential direction around the center axis to guide the first engaging portion in the insertion direction as the relative rotation phase transitions gradually to the engagement phase, and the first guide inclined surface makes contact with the first engaging portion, in a case where the relative rotation phase is the deformation phase, such that a pressing force that causes the elastic deformation is generated between the first guide inclined surface and the first engaging portion, while guiding the first engaging portion to the engagement phase.
8. The developer container according to claim 7, wherein the first engaging portion includes a first guided inclined surface that makes contact with the first guide inclined surface in a case where the relative rotation phase is the deformation phase, the first guided inclined surface extending in a direction inclined relative to both of the insertion direction and a circumferential direction around the center axis to guide the first engaging portion in the insertion direction as the relative rotation phase transitions gradually to the engagement phase.
9. The developer container according to claim 7, wherein the insertion portion includes an outer

circumferential surface that faces an inner circumferential surface of the through hole of the inserted portion, the first engaged portion is a groove portion provided to be recessed from the inner circumferential surface in a radial direction around the center axis, the first engaging portion is a protrusion provided on the outer circumferential surface protruding in the radial direction, the groove portion including: an insertion guide portion including an insertion hole open in an end face on an opposite side from the insertion direction of the connector member for allowing insertion of the protrusion in a case where the insertion portion is inserted into the inserted portion, and extending from the insertion hole in the insertion direction to guide the protrusion in the insertion direction; a deformation guide portion including the first guide inclined surface and extending in the circumferential direction from downstream in the insertion direction of the insertion guide portion to guide a movement of the protrusion, in a case where the relative rotation phase is the deformation phase; and an engagement retaining portion provided downstream of the deformation guide portion and including: a first circumferential restricting portion and a second circumferential restricting portion, which face the protrusion in the engagement phase respectively in a first direction and a second direction, which is opposite from the first direction, along the circumferential direction; and a counter-insertion direction restricting portion facing the protrusion in the insertion direction.

10. The developer container according to claim 7, wherein a set of the first engaging portion and the first engaged portion is provided at a plurality of positions spaced apart in a circumferential direction around the center axis, and at least one of the plurality of sets is positioned differently from other sets in the insertion direction.

11. The developer container according to claim 7, wherein the connector member includes a second guide inclined surface positioned downstream in the insertion direction relative to the second engaging portion and making contact with the first engaging portion in a case where the relative rotation phase is the deformation phase, the second guide inclined surface extending in a direction inclined relative to both of the insertion direction and the circumferential direction to guide the second engaging portion in a counter-insertion direction opposite from the insertion direction as the relative rotation phase transitions gradually to the engagement phase, and the second guide inclined surface makes contact with the second engaging portion, in a case where the relative rotation phase is the deformation phase, such that a pressing force that causes the elastic deformation is generated between the second guide inclined surface and the second engaging portion, while guiding the second engaging portion to the engagement phase.

12. The developer container according to claim 11, wherein the discharge path forming member includes a second guided inclined surface that is positioned closer to the center axis than the second engaged portion in a direction perpendicular to the center axis in a case where the relative rotation phase is the deformation phase, the second guided inclined surface extending in a direction perpendicular to the insertion direction and inclined relative to both of a radial direction around the center axis and the circumferential direction to guide the second engaged portion away from the center axis as the relative rotation phase transitions gradually to the engagement phase, and the second guided inclined surface makes contact with the second engaged portion, in a case where the relative rotation phase is the deformation phase, such that a pressing force that causes the elastic deformation is generated between the second guided inclined surface and the second engaged portion, while guiding the second engaged portion to the engagement phase.

13. The developer container according to claim 12, wherein the second engaged portion is a second protrusion protruding in a counter-insertion direction opposite from the insertion direction in which the insertion portion is inserted into the inserted portion, the second engaging portion is a third protrusion protruding in a direction intersecting the center axis and away from the center axis, the second protrusion includes: the second guide inclined surface; and a third circumferential restricting portion facing the third protrusion in the engagement phase in a first direction along the circumferential direction, and the third protrusion includes: the second guided inclined surface; and

a fourth circumferential restricting portion facing the second protrusion in the engagement phase in a second direction opposite from the first direction.
14. The developer container according to claim 1, further comprising a shutter that is provided movably to the discharge path forming member and that is able to open and close a discharge port of the discharge path in the discharge path forming member.
15. A method of manufacturing a developer container that contains developer to be supplied to a main body of an image forming apparatus, The developer container including: a container member including a container portion containing a developer, and an opening portion that opens the container portion; a connector member attached to the opening portion; and a discharge path forming member connected to the container member via the connector member and forming a discharge path for discharging the developer from the container portion, the connector member including an inserted portion having a through hole extending therethrough, and an engaged portion, the discharge path forming member including an insertion portion inserted into the through hole of the inserted portion, and an engaging portion provided correspondingly to the engaged portion, the engaged portion and the engaging portion being configured to come to a mutually engaged state through a process in which at least one of the connector member and the discharge path forming member is elastically deformed by a relative rotation between the connector member and the discharge path forming member about a predetermined rotation axis as a center axis, in a state where the insertion portion is inserted into the inserted portion, and to restrict, in the engaged state, a relative movement of the connector member and the discharge path forming member in a direction of the rotation axis, the method comprising: a first assembling step of attaching the connector member to the container member; a filling step of filling the container portion with the developer, and a second assembling step of connecting the discharge path forming member to the container member via the connector member.
16. The method of manufacturing a developer container according to claim 15, wherein, in the first assembling step, the connector member attached to the container member forms a fill port for the container portion to be filled with the developer in the filling step, and in the second assembling step, the discharge path forming member forms the discharge path in a case of being attached to the fill port formed by the connector member in the filling step.
17. A developer container comprising: a container portion having an opening portion at one end thereof and containing developer; a nozzle having an engaging portion, and a discharge port aligned with the container portion in a predetermined direction for discharging the developer contained in the container portion; and a coupling member having an engaged portion that engages with the engaging portion, and attached to the opening portion, and coupling the nozzle and the container portion, the engaged portion including a first restricting portion and a second restricting portion, the engaging portion being configured to move, in a case where the nozzle is moved in an intersecting direction intersecting the predetermined direction relative to the coupling member, from a first position to a second position in the intersecting direction relative to the engaged portion, the first position being a position where the nozzle is allowed to move in the predetermined direction relative to the coupling member, the second position being a position where the first restricting portion restricts a movement of the nozzle in the predetermined direction relative to the coupling member, and where the second restricting portion restricts a movement of the engaging portion relative to the engaged portion from the second position to the first position in the intersecting direction.
18. The developer container according to claim 17, wherein at least one of the engaging portion and the engaged portion is configured to be able to undergo an elastic deformation that allows the engaging portion to move to the second position by a contact between the engaging portion and the engaged portion in a case where the engaging portion moves from the first position to the second position.
19. The developer container according to claim 18, wherein the engaged portion includes a

deformation guide portion that induces the elastic deformation in at least one of the engaging portion and the engaged portion in a case where the engaging portion moves from the first position to the second position, and that makes sliding contact with the engaging portion so as to guide the engaging portion to the second position.